

The **dual problem** is

$$\begin{aligned} \text{DSDP} \quad d^* = \quad & \text{maximize} \quad \mathbf{b}^T \mathbf{y} \\ & \text{subject to} \quad T^{\text{adj}} \mathbf{y} \preceq C \\ & \text{(equivalently)} \quad T^{\text{adj}} \mathbf{y} + Z = C, \quad Z \succeq 0. \end{aligned}$$

Weak duality $p^* \geq d^*$ holds. If $p^* = d^*$ and d^* is attained, then **Strong duality** holds.

Complementary slackness holds if $\text{tr } ZX = 0$; **strict complementarity** holds if, in addition, $Z + X \succ 0$.

Facts:

1. The optimal strategy of player X over all possible strategies by the dual player Y has optimal value greater or equal to that of the optimal strategy for player Y over all strategies for player X , i.e.,

$$p^* = \min_{X \succeq 0} \max_{\mathbf{y}} L(X, \mathbf{y}) \geq d^* = \max_{\mathbf{y}} \min_{X \succeq 0} b^T \mathbf{y} + \text{tr } X(C - T^{\text{adj}} \mathbf{y}).$$

The first equality in the min–max problem can be seen from the *hidden constraint* for the inner maximization problem. The inequality follows from interchanging min and max. The hidden constraint in the inner minimization of the max–min problem yields our dual problem and weak duality. However, *strong duality* can fail. (See [RTW97].)

2. For $X, Z \succeq 0$, complementary slackness $\text{tr } ZX = 0$ is equivalent to $ZX = 0$.
3. *Characterization of Optimality Theorem*

The primal-dual pair $X, (\mathbf{y}, Z)$, with $X \succeq 0, Z \succeq 0$, is primal-dual optimal for SDP and DSDP if and only if

$$\begin{aligned} & T^{\text{adj}} \mathbf{y} + Z - C = 0 && \text{(dual feasibility)} \\ \text{(OPT)} \quad & TX - \mathbf{b} = 0 && \text{(primal feasibility)} \\ & ZX = 0 && \text{(complementary slackness).} \end{aligned}$$

4. The theorem in the previous fact is used to derive efficient primal-dual interior-point methods.
5. Again we note the similarity between the above optimality conditions and those for LP, where Z, X are diagonal matrices. However, in contrast to LP, the Goldman–Tucker theorem [GT56] can fail, i.e., a strict complementary solution may not exist. This means that there may be no optimal solution with $X + Z \succ 0$ positive definite; see [Sha00], [WW05]. This can result in loss of superlinear convergence for numerical algorithms for SDP. Moreover, in LP, Z, X are diagonal matrices and so is ZX . This means that the optimality conditions OPT are a square system. However, for SDP, the product of two symmetric matrices ZX is not necessarily symmetric. Therefore, the optimality conditions are an overdetermined nonlinear system of equations. This gives rise to many subtle algorithmic difficulties.

Examples:

1. The lack of closure noted in Fact 4 in section 51.3 is illustrated by the sequence

$$\begin{bmatrix} \frac{1}{7} & 1 \\ 1 & i \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & -i \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \notin \text{PSD} + \text{Span} \left\{ \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}.$$

This gives rise to an SDP with a duality gap. Let $C = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $A_1 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$. Then a primal SDP is $0 = \min \text{tr } CX$ s.t. $\text{tr } A_1 X = 0, X \succeq 0$. But the dual program has optimal value $-\infty = \max 0\mathbf{y}$ s.t. $\mathbf{y}A_1 \preceq C$, i.e., it is infeasible.

2. For fixed $\alpha > 0$, we use parameters $\mathbf{b} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $C = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $A_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $A_2 =$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$. Then we get the primal and dual SDP pair with a finite duality gap: $\alpha = \min \alpha U_{11}$ s.t.

$$U_{22} = 0, U_{11} + 2U_{23} = 1, U \succeq 0; 0 = \max y_2 \text{ s.t. } \begin{bmatrix} y_2 & 0 & 0 \\ 0 & y_1 & y_2 \\ 0 & y_2 & 0 \end{bmatrix} \preceq \begin{bmatrix} \alpha & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

3. We now verify that $X^* = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}$ is optimal for MCSDP in Example 1 in Section 51.1.

The dual of MCSDP is $\min e^T \mathbf{y}$ s.t. $\text{Diag}(\mathbf{y}) \succeq \frac{1}{4}L$. Since $\mathbf{y} = [1 \ 1 \ 2]^T$ is feasible for this dual, optimality follows from checking strong duality $\text{tr} LX^* = e^T \mathbf{y} = 4$. Equivalently, one can check complementary slackness $(\text{Diag}(\mathbf{y}) - L)X^* = 0$.

51.5 Strong Duality without a Constraint Qualification

Definitions:

A **constraint qualification**, CQ, is a condition on the constraints that guarantees that strong duality holds at an optimum point.

Slater's Constraint Qualification (strict feasibility) is defined as: $\exists X \succ 0, TX = \mathbf{b}$.

Facts:

1. Suppose that Slater's CQ holds. Then strong duality holds for SDP and DSDP, i.e., $p^* = d^*$ and d^* is attained.
2. [BW81] *Duality Theorem*: Suppose that p^* is finite. Let $\mathcal{F}_e$ denote the minimal face of PSD that contains the feasible set of SDP. Consider the extended dual program

$$\begin{aligned} d_e^* = \quad & \text{maximize} && \mathbf{b}^T \mathbf{y} \\ \text{DSDP}_e \quad & \text{subject to} && T^{\text{adj}} \mathbf{y} + Z = C \\ & && Z \in \mathcal{F}_e^+. \end{aligned}$$

Then $p^* = d_e^*$ and d_e^* is attained, i.e., strong duality holds between SDP and DSDP_e.

3. An algorithm for finding the minimal face $\mathcal{F}_e$ defined above is given in [BW81].

Examples:

We can apply the strong duality results to the two examples given in Section 51.4.

1. For the first example, replace PSD in the primal with the minimal face $\mathcal{F} = \text{PSD} \cap \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^\perp = \text{cone}\{A_1\}$; so $\mathcal{F}^+ = \{A_1\}^+$ and the dual is now feasible.
2. For the second example, replace PSD in the primal with the minimal face $\mathcal{F} = \text{PSD} \cap \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}^\perp$; so U_{23} is no longer restricted to be 0 in the dual.
3. The strong duality results are needed in many SDP applications to combinatorial problems where Slater's constraint qualification (strict feasibility) fails, e.g., [WZ99], [ZKR98].

51.6 A Primal-Dual Interior-Point Algorithm

As mentioned above, it was the development of efficient polynomial-time algorithms for SDP at the end of the 1980s that spurred the increased interest in the field. We look at the specific method derived in [HRV96]. (See also [KSH97], [Mon97].)

We assume that Slater's constraint qualification (strict feasibility) holds for both primal and dual problems.

Definitions:

For each $\mu > 0$, the log-barrier approach applied to DSDP requires the solution of the **log-barrier problem**

$$\begin{aligned} \text{DSDP}_\mu \quad d_\mu^* = \quad & \text{maximize} \quad \mathbf{b}^T \mathbf{y} + \mu \log \det Z \\ & \text{subject to} \quad T^{\text{adj}} \mathbf{y} + Z = C, \quad Z \succ 0. \end{aligned}$$

The **Lagrangian** is $L(X, \mathbf{y}, Z) = \mathbf{b}^T \mathbf{y} + \mu \log \det Z + \text{tr } X(C - T^{\text{adj}} \mathbf{y} - Z)$, where we use X for the Lagrange multiplier vector.

Facts:

1. The derivative, with respect to $(X, \mathbf{y}, Z)$ of the Lagrangian, yields the optimality conditions for the barrier problem:

$$\begin{aligned} \text{OPT}_\mu \quad & T^{\text{adj}} \mathbf{y} + Z - C = 0 && \text{(dual feasibility)} \\ & TX - b = 0 && \text{(primal feasibility)} \\ & \mu Z^{-1} - X = 0 && \text{(perturbed complementary slackness).} \end{aligned}$$

We let $X_\mu, \mathbf{y}_\mu, Z_\mu$ denote the unique solution of OPT_μ , i.e., the unique optimum of the log-barrier problem. The set $\{X_\mu, \mathbf{y}_\mu, Z_\mu : \mu \downarrow 0\}$ is called the **central path**. Successful primal-dual interior-point methods are actually path-following methods, i.e., they follow the central path to the optimum at $\mu = 0$.

2. The last equation in the optimality conditions OPT_μ is very nonlinear and leads to ill-conditioning as μ gets close to zero. Therefore, it is replaced by the more familiar

$$\text{OPT}_\mu \quad ZX - \mu I = 0 \quad \text{(perturbed complementary slackness).}$$

We denote the resulting optimality conditions using

$$F_\mu(X, \mathbf{y}, Z) = 0.$$

This system has the same appearance as the one used in LP. However, it is an overdetermined nonlinear system, i.e., $F_\mu : \mathcal{S}_n \times \mathbb{R}^m \times \mathcal{S}_n \rightarrow \mathcal{S}_n \times \mathbb{R}^m \times \mathbb{R}^{n \times n}$, since the product ZX is not necessarily symmetric. Therefore, the successful application of Newton's method done in LP must be modified. A Gauss-Newton method is used in [KMR01]. Another approach is to symmetrize the last equation of OPT_μ . A general symmetrization scheme is presented in [MZ98].

3. We derive and present (arguably) the most popular algorithm for SDP, the **HKM method** [HRV96], [KSH97], [Mon97].

We consider a current estimate $X, \mathbf{y}, Z$ of the optimum of the log-barrier problem DSDP_μ , where both $X \succ 0$, $Z \succ 0$. Since we want $ZX = \mu I$, we set $\mu = \frac{1}{n} \text{tr } ZX$. We use a centering parameter $0 < \sigma$ that is adaptive in that it decreases (respectively increases) according to the decrease (respectively increase) in the duality gap estimate at each iteration. Small σ signifies an aggressive step in decreasing μ to 0. We replace $\mu \leftarrow \sigma \mu$ in the optimality conditions. We denote

the optimality conditions using $F_\mu = F_\mu(X, \mathbf{y}, Z) = \begin{bmatrix} R_d \\ \mathbf{r}_p \\ R_c \end{bmatrix}$, i.e., using three residuals. To derive the HKM method, we solve for the Newton direction $d_N = \begin{bmatrix} \Delta X \\ \Delta \mathbf{y} \\ \Delta Z \end{bmatrix}$ in the Newton equation

$$F'_\mu(X, \mathbf{y}, Z) = \begin{bmatrix} 0 & T^{\text{adj.}} & I \\ T & 0 & 0 \\ Z & 0 & \cdot X \end{bmatrix} d_N = - \begin{bmatrix} R_d \\ \mathbf{r}_p \\ R_c \end{bmatrix} = -F_\mu.$$

To solve the above linearized system efficiently, we use block elimination. We use the first equation to solve for

$$\Delta Z = -R_d - T^{\text{adj.}} \Delta \mathbf{y}.$$

We then substitute into the third equation and solve for

$$\Delta X = Z^{-1} ((R_d + T^{\text{adj.}} \Delta \mathbf{y})X - R_c).$$

We then substitute this into the second equation and obtain the so-called *Schur complement* system, which is similar to the normal equation in LP:

$$\begin{aligned} TZ^{-1}T^{\text{adj.}}\Delta \mathbf{y}X &= -TZ^{-1}(R_dX - R_c) - \mathbf{r}_p \\ &= -TZ^{-1}(R_dX - ZX + \mu I) - TX + \mathbf{b} \\ &= \mathbf{b} - TZ^{-1}(R_dX + \mu I). \end{aligned}$$

This is a positive definite linear system. Once $\Delta \mathbf{y}$ is found, we can backsolve for $\Delta X, \Delta Z$.

We now observe another major difference with LP and a major difficulty for SDP. Though ΔZ is now symmetric, this is not true, in general, for ΔX , since the product of symmetric matrices is not necessarily symmetric. Therefore, we symmetrize $\Delta X \leftarrow \frac{1}{2}(\Delta X + \Delta X^T)$. We have now obtained the search direction $(\Delta X, \Delta \mathbf{y}, \Delta Z)$.

Next, a line search is performed that maintains X, Z positive definite, i.e., we find the next iterate $(X, \mathbf{y}, Z)$ using the primal and dual steplengths $X \leftarrow X + \alpha_p \Delta X \succ 0, Z \leftarrow Z + \alpha_d \Delta Z \succ 0$ and set $\mathbf{y} \leftarrow \mathbf{y} + \alpha_d \Delta \mathbf{y}$. We then update σ, μ and repeat the iteration, i.e., we go back to Fact 3 above.

4. See [HRV96] for more details on the HKM algorithm, including a predictor-corrector approach. Details on different symmetrization schemes that lead to various search directions and convergence proofs can be found in [MT00]. Algorithms based on bundle methods that handle large-scale SDP can be found in [HO00].

51.7 Applications of SDP

There are a surprising number of important applications for SDP, several of which have already been mentioned. Early applications in engineering involved solutions of linear matrix inequalities, LMIs, e.g., Lyapunov and Riccati equations; see [BEF94]. Semidefinite programming plays an important role in combinatorial optimization where it is used to solve nonlinear convex relaxations of NP-hard problems [Ali95], [WZ99], [ZKR98]. This includes strong theoretical results that characterize the quality of the bounds; see [GR00]. Matrix completion problems have been mentioned above. This includes Euclidean Distance Matrix completion problems with applications to, e.g., molecular conformation; see [AKW99]. Further applications to control theory, finance, nonlinear programming, etc. can be found in [BEF94] and [WSV00].

Facts:

1. SDP arises naturally when finding relaxations for hard combinatorial problems. We saw one simple case, the Max-Cut problem, MC, in the introduction in Example 1, i.e., MC was modeled as a quadratic maximization problem and the binary ± 1 variables were modeled using the quadratic constraints $x_j^2 = 1$.
2. In addition to ± 1 binary variables in Fact 1, we can model 0,1 variables using the quadratic constraints $x_j^2 - x_j = 0$. Linear constraints $A\mathbf{x} - \mathbf{b} = 0$ can be modeled using the quadratic constraint $\|A\mathbf{x} - \mathbf{b}\|^2 = 0$. Thus, in general, many hard combinatorial optimization problems are equivalent to a quadratically constrained quadratic program, QQP. Denote the quadratic function $q_i(\mathbf{y}) = \frac{1}{2}\mathbf{y}^T Q_i \mathbf{y} + \mathbf{y}^T b_i + c_i, \mathbf{y} \in \mathbb{R}^n, i = 0, 1, \dots, m$. Then we have

$$\begin{aligned} \text{(QQP)} \quad q^* = \quad & \min \quad q_0(\mathbf{y}) \\ & \text{subject to} \quad q_i(\mathbf{y}) = 0, \quad i = 1, \dots, m. \end{aligned}$$

(For simplicity, we restrict to equality constraints.)

3. SDP arises naturally from the Lagrangian relaxation of QQP in Fact 2 above. We write the Lagrangian $L(\mathbf{y}, \mathbf{x})$, with Lagrange multiplier vector $\mathbf{x}$.

$$L(\mathbf{y}, \mathbf{x}) = \begin{array}{lll} \text{quadratic in } \mathbf{y} & \text{linear in } \mathbf{y} & \text{constant in } \mathbf{y} \\ \frac{1}{2}\mathbf{y}^T(Q_0 - \sum_{i=1}^m x_i Q_i)\mathbf{y} & + \mathbf{y}^T(b_0 - \sum_{i=1}^m x_i b_i) & + (c_0 - \sum_{i=1}^m x_i c_i). \end{array}$$

Weak duality follows from the primal-dual pair relationship

$$q^* = \min_{\mathbf{y}} \max_{\mathbf{x}} L(\mathbf{y}, \mathbf{x}) \geq d^* = \max_{\mathbf{x}} \min_{\mathbf{y}} L(\mathbf{y}, \mathbf{x}).$$

We now homogenize the Lagrangian by adding y_0 to the linear term: $y_0 y^T(b_0 - \sum_{i=1}^m x_i b_i), y_0^2 = 1$. Then, after moving the constraint on y_0 into the Lagrangian, the dual program becomes

$$\begin{aligned} d^* &= \max_{\mathbf{x}} \min_{\mathbf{y}} L(\mathbf{y}, \mathbf{x}) \\ &= \max_{\mathbf{x}, t} \min_{\mathbf{y}} \quad \frac{1}{2}\mathbf{y}^T(Q_0 - \sum_{i=1}^m x_i Q_i)\mathbf{y} + t y_0^2 \\ &\quad + y_0 \mathbf{y}^T(b_0 - \sum_{i=1}^m x_i b_i) \\ &\quad + (c_0 - \sum_{i=1}^m x_i c_i)(-t). \end{aligned}$$

We now exploit the *hidden constraint* from the inner minimization of a homogeneous quadratic function, i.e., the Hessian of the Lagrangian must be positive semidefinite. This gives rise to the SDP

$$\begin{aligned} d^* &= \sup \quad -t - \sum_{i=1}^m x_i c_i \\ \text{(D)} \quad & \text{s.t.} \quad T \begin{bmatrix} t \\ \mathbf{x} \end{bmatrix} \leq B \\ & \quad \mathbf{x} \in \mathbb{R}^m, \quad t \in \mathbb{R}, \end{aligned}$$

where the linear transformation $T : \mathbb{R}^{m+1} \rightarrow \mathcal{S}_{n+1}$ and the matrix B are defined by

$$B = \begin{bmatrix} 0 & b_0^T \\ b_0 & Q_0 \end{bmatrix}, \quad T \begin{bmatrix} t \\ \mathbf{x} \end{bmatrix} = \begin{bmatrix} -t & \sum_{i=1}^m x_i b_i^T \\ \sum_{i=1}^m x_i b_i & \sum_{i=1}^m x_i Q_i \end{bmatrix}.$$

The dual, DD, of the program D gives rise to the SDP relaxation of QQP.

$$\begin{aligned}
 p^* = \inf \quad & \text{tr } BU \\
 (\text{DD}) \quad & \text{s.t. } T^{\text{adj}} U = \begin{bmatrix} -1 \\ -c \end{bmatrix} \\
 & U \succeq 0.
 \end{aligned}$$

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Random Vectors and Linear Statistical Models

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McGill University

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52.1 Introduction and Mise-En-Scène

Linear algebra is used extensively in statistical science, in particular in linear statistical models, see, e.g., Graybill [Gra01], Ravishanker and Dey [RD02], Rencher [Ren00], Searle [Sea97], Seber and Lee [SL03], Sengupta and Jammalamadaka [SJ03], and Stapleton [Sta95]; as well as in applied economics, see, e.g., Searle and Willett [SW01]; econometrics, see, e.g., Davidson and MacKinnon [DM04], Magnus and Neudecker [MN99], and Rao and Rao [RR98]; Markov chain theory, see, e.g., Chapter 54 or Kemeny and Snell [KS83]; multivariate statistical analysis, see, e.g., Anderson [And03], Kollo and von Rosen [KvR05], and Seber [Seb04]; psychometrics, see, e.g., Takane [Tak04] and Takeuchi, Yanai, and Mukherjee [TYM82]; and random matrix theory, see, e.g., Bleher and Its [BI01] and Mehta [Meh04]. Moreover, there are several books on linear algebra and matrix theory written by (and mainly for) statisticians, see, e.g., Bapat [Bap00], Graybill [Gra83], Hadi [Had96], Harville [Har97], [Har01], Healy [Hea00], Rao and Rao [RR98], Schott [Sch05], Searle [Sea82], and Seber [Seb06].

As Miller and Miller [MM04, p. 1] and Wackerly, Mendenhall, and Scheaffer [WMS02, p. 2] point out: “Statistics is a theory of information, with inference making as its objective” and may be viewed as encompassing “the science of basing inferences on observed data and the entire problem of making decisions in the face of uncertainty.”

In this chapter we present an introduction (Section 52.2) to the use of linear algebra in studying properties of random vectors (Section 52.3) and in linear statistical models (Section 52.4); for further uses, see, e.g., [PSS05] and [PS05], and for an introduction to the use of linear algebra in multivariate statistical analysis see Chapter 53 or [PSS05].

We begin (in Section 52.2) with a brief introduction to the basic concepts of statistics and random variables [MM04], [WMS02], [WS00], with special emphasis (in Section 52.3) on random vectors — vectors where each entry is a random variable.

All matrices (with at least two rows and two columns) considered in this chapter are nonrandom.

52.2 Introduction to Statistics and Random Variables

Notation:

In this chapter most uppercase, light-face italic letters, in particular X , Y , and Z , denote scalar random variables, but the notation $P(A)$ is reserved for the probability of the set A . Throughout this chapter, the uppercase, light-face roman letter E denotes expectation.

Definitions:

The focus in statistics is on making inferences concerning a large body of data called a **population** based on a subset collected from it called a **sample**. An **experiment** associated with this sample is repeatable and its outcome is not predetermined. A **simple event** is one associated with an experiment, which cannot be decomposed, and a simple event corresponds to one and only one **sample point**. The **sample space** associated with an experiment is the set of all possible sample points.

Suppose that S is a sample space associated with an experiment. To every subset A of S we assign a real number $P(A)$, called the **probability** of A , so that the following axioms hold: (1) $P(A) \geq 0$, (2) $P(S) = 1$, (3) If $A_1, A_2, A_3, \dots$ form a sequence of pairwise mutually exclusive subsets in S , i.e., $A_i \cap A_j = \emptyset$ if $i \neq j$, then $P(A_1 \cup A_2 \cup A_3 \cup \dots) = \sum_{i=1}^{\infty} P(A_i)$.

A **random variable** is a real-valued function for which the domain is a sample space.

A random variable which can assume a finite or countably infinite number of values is **discrete**.

The **probability** $P(Y = y)$ that the random variable Y takes on the value y is defined as the sum of the probabilities of all sample points in S that have the value y , and the **probability function** of Y is the set of all the probabilities $P(Y = y)$.

If the random variable Y has the probability function $P(Y = 0) = p$ and $P(Y = 1) = 1 - p$ for some real number p in the interval $[0, 1]$, then Y is a **Bernoulli random variable**.

The **cumulative distribution function** (cdf) is $F(y) = P(Y \leq y)$ of the random variable Y for $-\infty < y < \infty$. A random variable Y is **continuous** when its cdf $F(y)$ is continuous for $-\infty < y < \infty$, and then its **probability density function** (pdf) $f(y) = dF(y)/dy$.

Suppose that the continuous random variable Y has pdf $f(y) = 1$ for $0 \leq y \leq 1$ and $f(y) = 0$ otherwise. Then Y follows a **uniform distribution** on the interval $[0, 1]$.

The **expectation** (or **expected value** or **mean** or **mean value**) $E(Y)$ of the random variable Y is $E(Y) = \sum_y yP(Y = y)$ when Y is discrete with probability function $P(Y = y)$ and is $E(Y) = \int_{-\infty}^{+\infty} yf(y)dy$ when Y is continuous with pdf $f(y)$.

The **variance** σ^2 of the random variable Y is

$$\sigma^2 = \text{var}(Y) = E((Y - \mu)^2),$$

where $\mu = E(Y)$ and the **standard deviation** $\sigma = \sqrt{\sigma^2}$.

Facts:

1. The variance $\sigma^2 = \text{var}(Y) = E(Y^2) - E^2(Y)$.
2. For any random variable Y , the expectation of its square $E(Y^2) \geq E^2(Y)$ with equality if and only if the random variable $Y = E(Y)$ with probability 1.
3. If Y is a Bernoulli random variable with probability function $P(Y = 0) = p$ and $P(Y = 1) = 1 - p$, then the expectation $E(Y) = 1 - p$ and the variance $\text{var}(Y) = p(1 - p)$.
4. If the random variable Z follows a uniform distribution on the interval $[0, 1]$, then the expectation $E(Z) = 1/2$ and the variance $\text{var}(Z) = 1/12$.

Examples:

1. Every person's blood type is A, B, AB, or O. In addition, each individual either has the Rhesus (Rh) factor (+) or does not (-). A medical technician records a person's blood type and Rh factor. The sample space for this experiment is $\{A+, B+, AB+, O+, A-, B-, AB-, O-\}$ with eight sample points.
2. Consider the experiment of tossing a single fair coin and define the random variable $Y = 0$ if the outcome is "heads," and $Y = 1$ if the outcome is "tails." Then Y is a Bernoulli random variable, and $P(Y = 0) = \frac{1}{2} = P(Y = 1)$, $E(Y) = \frac{1}{2}$, $\text{var}(Y) = \frac{1}{4}$.
3. Suppose that a bus always arrives at a particular stop in the interval between 12 noon and 1 p.m. and that the probability that the bus will arrive in any given subinterval of time is proportional only to the length of the subinterval. Let Y denote the length of time that a person arriving at the stop at 12 noon must wait for the bus to arrive, and let us code 12 noon as 0 and measure the time in hours. Then the random variable Y follows a uniform distribution on the interval $[0, 1]$.

52.3 Random Vectors: Basic Definitions and Facts

Linear algebra is extensively used in the study of random vectors, where we consider the simultaneous behavior of two or more random variables assembled as a vector. In this section all vectors and matrices are real.

Notation:

In this section uppercase, light-face italic letters, such as X , Y , and Z , denote scalar random variables and lowercase bold roman letters, such as $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$, denote random vectors. Uppercase, light-face italic letters such as A and B denote nonrandom matrices.

Definitions:

Let $A \in \mathbb{R}^{n \times k}$ and $B \in \mathbb{R}^{n \times q}$. Then the **partitioned matrix** $[A \mid B]$ is the $n \times (k + q)$ matrix formed by placing A next to B .

A $k \times 1$ **random vector** $\mathbf{y}$ is a vector $\mathbf{y} = [Y_1, \dots, Y_k]^T$ of k random variables $Y_1, \dots, Y_k$. The **expectation** (or **expected value** or **mean vector**) of $\mathbf{y}$ is the $k \times 1$ vector $E(\mathbf{y}) = [E(Y_1), \dots, E(Y_k)]^T$. Sometimes, for clarity, a vector of constants (belonging to $\mathbb{R}^n$) is called a **nonrandom vector** and a matrix of constants a **nonrandom matrix**. (Random matrices are not considered in this chapter.)

The **covariance** $\text{cov}(Y, Z)$ between the two random variables Y and Z is $\text{cov}(Y, Z) = E((Y - \mu)(Z - \nu))$, where $\mu = E(Y)$ and $\nu = E(Z)$.

The **correlation** (or **correlation coefficient** or **product-moment correlation**) $\text{cor}(Y, Z)$ between the two random variables Y and Z is $\text{cor}(Y, Z) = \text{cov}(Y, Z) / \sqrt{\text{var}(Y)\text{var}(Z)}$.

The **covariance matrix** (or **variance-covariance matrix** or **dispersion matrix**) of the $k \times 1$ random vector $\mathbf{y} = [Y_1, \dots, Y_k]^T$ is the $k \times k$ matrix $\text{var}(\mathbf{y}) = \Sigma$ of variances and covariances of all the entries of $\mathbf{y}$:

$$\begin{aligned} \text{var}(\mathbf{y}) = \Sigma &= [\sigma_{ij}] = [\text{cov}(Y_i, Y_j)] = [E(Y_i - \mu_i)(Y_j - \mu_j)] \\ &= E((\mathbf{y} - \boldsymbol{\mu})(\mathbf{y} - \boldsymbol{\mu})^T), \end{aligned}$$

where $\boldsymbol{\mu} = E(\mathbf{y})$. The determinant $\det \Sigma$ is the **generalized variance** of the random vector $\mathbf{y}$. The variances σ_{ii} are often denoted as σ_i^2 and, in this, chapter, we will assume that they are all positive. If $\sigma_i = 0$, then the random variable $Y_i = E(Y_i)$ with probability 1, and then we interpret Y_i as a constant. In statistics it is quite common to denote **standard deviations** as σ_i . (The reader should note that in all the other chapters of this book except in the two statistics chapters, σ_i denotes the i th largest singular value.)

The **cross-covariance matrix** $\text{cov}(\mathbf{y}, \mathbf{z})$ between the $k \times 1$ random vector $\mathbf{y} = [Y_1, \dots, Y_k]^T$ and the $q \times 1$ random vector $\mathbf{z} = [Z_1, \dots, Z_q]^T$ is the $k \times q$ matrix of all the covariances $\text{cov}(Y_i, Z_j); i = 1, \dots, k$ and $j = 1, \dots, q$:

$$\text{cov}(\mathbf{y}, \mathbf{z}) = [\text{cov}(Y_i, Z_j)] = [E(Y_i - \mu_i)(Z_j - \nu_j)] = E((\mathbf{y} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\nu})^T),$$

where $\boldsymbol{\mu} = [\mu_i] = E(\mathbf{y})$ and $\boldsymbol{\nu} = [\nu_j] = E(\mathbf{z})$. The random vectors $\mathbf{y}$ and $\mathbf{z}$ are **uncorrelated** whenever the cross-covariance matrix $\text{cov}(\mathbf{y}, \mathbf{z}) = \mathbf{0}$.

The **correlation matrix** $\text{cor}(\mathbf{y}) = R$, say, of the $k \times 1$ random vector $\mathbf{y} = [Y_1, \dots, Y_k]^T$, is the $k \times k$ matrix of correlations of all the entries in $\mathbf{y}$: $\text{cor}(\mathbf{y}) = R = [\rho_{ij}] = [\text{cor}(Y_i, Y_j)] = [\frac{\sigma_{ij}}{\sigma_i \sigma_j}]$, where $\sigma_i = \sqrt{\sigma_{ii}} =$ standard deviation of Y_i ; $\sigma_i, \sigma_j > 0$.

Let $\mathbf{1}_k$ denote the $k \times 1$ column vector with every entry equal to 1. Then $J_k = \mathbf{1}_k \mathbf{1}_k^T$ is the $k \times k$ **all-ones matrix** (with all k^2 entries equal to 1) and $C_k = I_k - \frac{1}{k} J_k$ is the $k \times k$ **centering matrix**.

Suppose that the real positive numbers $p_1, p_2, \dots, p_k$ are such that $\sum_{i=1}^k p_i = 1$. Then the $k \times 1$ random vector $\mathbf{y} = [Y_1, \dots, Y_k]^T$ follows a **multinomial distribution** with parameters n and $p_1, p_2, \dots, p_k$ if the joint probability function of $Y_1, Y_2, \dots, Y_k$ is given by

$$P(Y_1 = y_1, Y_2 = y_2, \dots, Y_k = y_k) = \frac{n!}{y_1! y_2! \dots y_k!} p_1^{y_1} p_2^{y_2} \dots p_k^{y_k},$$

where for each i , $y_i = 0, 1, 2, \dots, n$ and $\sum_{i=1}^k y_i = n$. When $k = 2$ the distribution is **binomial**, and when $k = 3$ the distribution is **trinomial**.

Let the symmetric matrices $A \in \mathbb{R}^{k \times k}$ and $B \in \mathbb{R}^{k \times k}$. Then $A \succeq B$ means $A - B$ is positive semidefinite and $A \succ B$ means $A - B$ is positive definite. The partial ordering induced by $\succeq$ is called the **partial semidefinite ordering** (or **Loewner partial ordering** or **Löwner partial ordering**). (See Section 8.5 for more information.)

Let the $(k + q) \times 1$ random vector $\mathbf{x}$ have covariance matrix Σ . Consider the following partitioning:

$$\mathbf{x} = \begin{bmatrix} \mathbf{y} \\ \mathbf{z} \end{bmatrix}, \quad E(\mathbf{x}) = \begin{bmatrix} \boldsymbol{\mu} \\ \boldsymbol{\nu} \end{bmatrix}, \quad \text{var}(\mathbf{x}) = \Sigma = \begin{bmatrix} \Sigma_{yy} & \Sigma_{yz} \\ \Sigma_{zy} & \Sigma_{zz} \end{bmatrix},$$

where $\mathbf{y}$ and $\mathbf{z}$ have k and q elements, respectively. Then the **partial covariance matrix** $\Sigma_{zz \cdot y}$ of the $q \times 1$ random vector $\mathbf{z}$ after **adjusting for** (or **controlling for** or **removing the effect of** or **allowing for**) the $k \times 1$ random vector $\mathbf{y}$ is the (uniquely defined) **generalized Schur complement**

$$\Sigma / \Sigma_{yy} = [\sigma_{ij \cdot y}] = \Sigma_{zz} - \Sigma_{zy} \Sigma_{yy}^{-1} \Sigma_{yz}$$

of Σ_{yy} in Σ ; any **generalized inverse** Σ_{yy}^{-} satisfying $\Sigma_{yy} = \Sigma_{yy} \Sigma_{yy}^{-} \Sigma_{yy}$ may be chosen.

When Σ_{yy} is positive definite, we use the inverse Σ_{yy}^{-1} instead of a generalized inverse Σ_{yy}^{-} and refer to $\Sigma / \Sigma_{yy} = \Sigma_{zz} - \Sigma_{zy} \Sigma_{yy}^{-1} \Sigma_{yz}$ as the **Schur complement** of Σ_{yy} in Σ .

The $q \times 1$ random vector $\mathbf{e}_{z \cdot y} = \mathbf{z} - \boldsymbol{\nu} - \Sigma_{zy} \Sigma_{yy}^{-1} (\mathbf{y} - \boldsymbol{\mu})$ is the (uniquely defined) vector of **residuals** of the $q \times 1$ random vector $\mathbf{z}$ from its **regression** on the $k \times 1$ random vector $\mathbf{y}$.

The ij th entry of the **partial correlation matrix** of $q \times 1$ random vector $\mathbf{z} = [Z_1, \dots, Z_q]^T$ after adjusting for the $k \times 1$ random vector $\mathbf{y}$ is the **partial correlation coefficient** between z_i and z_j after adjusting for $\mathbf{y}$:

$$\rho_{ij \cdot y} = \frac{\sigma_{ij \cdot y}}{\sqrt{\sigma_{ii \cdot y} \sigma_{jj \cdot y}}}; \quad i, j = 1, \dots, q,$$

which is well defined provided the diagonal entries of the associated partial covariance matrix are all positive.

Facts:

1. [WMS02, Th. 5.13, p. 265] Let the $k \times 1$ random vector $\mathbf{y} = [Y_1, \dots, Y_k]^T$ follow a multinomial distribution with parameters n and $p_1, \dots, p_k$ and let the $k \times 1$ vector $\mathbf{p} = [p_1, \dots, p_k]^T$. Then:
 - The random variable Y_i can be represented as the sum of n independently and identically distributed Bernoulli random variables with parameter p_i ; $i = 1, \dots, k$.
 - The expectation $E(\mathbf{y}) = n\mathbf{p}$ and the covariance matrix

$$\text{var}(\mathbf{y}) = n(\text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^T) = \Psi_k,$$

say, where $\text{diag}(\mathbf{p})$ is the $k \times k$ diagonal matrix formed from the $k \times 1$ nonrandom vector $\mathbf{p}$.

- The covariance matrix Ψ_k is singular since all its row (and column) totals are 0, and the $\text{rank}(\Psi_k) = k - 1$.
2. When the $k \times 1$ multinomial probability vector $\mathbf{p} = \mathbf{1}_k/k$, then the multinomial covariance matrix $\Psi_k = \frac{n}{k}C_k$, where C_k is the $k \times k$ centering matrix.
 3. The $k \times k$ covariance matrix $\text{var}(\mathbf{y}) = \Sigma = E(\mathbf{y}\mathbf{y}^T) - \boldsymbol{\mu}\boldsymbol{\mu}^T$, where $\boldsymbol{\mu} = E(\mathbf{y})$.
 4. The $k \times k$ correlation matrix $\text{cor}(\mathbf{y}) = [\text{diag}(\Sigma)]^{-1/2}\Sigma[\text{diag}(\Sigma)]^{-1/2}$, where $\Sigma = \text{var}(\mathbf{y})$.
 5. The $k \times q$ cross-covariance matrix

$$\text{cov}(\mathbf{y}, \mathbf{z}) = \Sigma_{yz} = \Sigma_{zy}^T = (\text{cov}(\mathbf{z}, \mathbf{y}))^T = E(\mathbf{y}\mathbf{z}^T) - \boldsymbol{\mu}\boldsymbol{\nu}^T,$$

where $\boldsymbol{\mu} = E(\mathbf{y})$ and $\boldsymbol{\nu} = E(\mathbf{z})$.

6. [RM71, Lemma 2.2.4, p. 21]: The product $AB^{-1}C$ (for $A \neq \mathbf{0}$, $C \neq \mathbf{0}$) is invariant with respect to the choice of $B^{-1} \iff \text{range}(C) \subseteq \text{range}(B)$ and $\text{range}(A^T) \subseteq \text{range}(B^T)$.
7. Consider the $(k + q) \times (k + q)$ covariance matrix

$$\Sigma = \begin{bmatrix} \Sigma_{yy} & \Sigma_{yz} \\ \Sigma_{zy} & \Sigma_{zz} \end{bmatrix}.$$

Then the range (or column space) $\text{range}(\Sigma_{yz}) \subseteq \text{range}(\Sigma_{yy})$ and $\text{range}(\Sigma_{zy}) \subseteq \text{range}(\Sigma_{zz})$, and, hence, the matrix $\Sigma_{zy}\Sigma_{yy}^{-}\Sigma_{yz}$ and the generalized Schur complement $\Sigma/\Sigma_{yy} = \Sigma_{zz} - \Sigma_{zy}\Sigma_{yy}^{-}\Sigma_{yz}$ are invariant (unique) with respect to the choice of generalized inverse Σ_{yy}^{-} .

8. Let the $k \times 1$ random vector $\mathbf{x} = [X_1, \dots, X_k]^T$. Then the $k \times 1$ centered vector $C_k\mathbf{x} = [X_1 - \bar{X}, \dots, X_k - \bar{X}]^T$, where C_k is the $k \times k$ centering matrix and the arithmetic mean (or average) $\bar{X} = \sum_{i=1}^k X_i/k$.
9. A nonsingular positive semidefinite matrix is positive definite.
10. A covariance matrix is always symmetric and positive semidefinite.
11. A cross-covariance matrix is usually rectangular.
12. A correlation matrix is always symmetric and positive semidefinite.
13. The diagonal entries of a correlation matrix are all equal to 1 and the off-diagonal entries are all at most equal to 1 in absolute value.
14. [PS05, p. 168] The (generalized) Schur complement of the leading principal submatrix of a positive semidefinite matrix is positive semidefinite.
15. Let $\mathbf{y}$ be a $k \times 1$ random vector with covariance matrix Σ . Then the variance $\text{var}(\mathbf{a}^T\mathbf{y}) = \mathbf{a}^T\Sigma\mathbf{a}$ for all nonrandom $\mathbf{a} \in \mathbb{R}^k$. (Since variance must be nonnegative this fact shows that a covariance matrix must be positive semidefinite.)
16. Let $\mathbf{y}$ be a $k \times 1$ random vector with expectation $\boldsymbol{\mu} = E(\mathbf{y})$, covariance matrix $\Sigma = \text{var}(\mathbf{y})$, and let the matrix $A \in \mathbb{R}^{n \times k}$ and the nonrandom vector $\mathbf{b} \in \mathbb{R}^n$. Then the expectation $E(A\mathbf{y} + \mathbf{b}) = AE(\mathbf{y}) + \mathbf{b} = A\boldsymbol{\mu} + \mathbf{b}$ and the covariance matrix $\text{var}(A\mathbf{y} + \mathbf{b}) = A\text{var}(\mathbf{y})A^T = A\Sigma A^T$.
17. Let $\mathbf{y}$ be a $k \times 1$ random vector with expectation $\boldsymbol{\mu} = E(\mathbf{y})$, covariance matrix $\Sigma = \text{var}(\mathbf{y})$, and let the matrix $A \in \mathbb{R}^{k \times k}$, not necessarily symmetric. Then $E(\mathbf{y}^T A \mathbf{y}) = \boldsymbol{\mu}^T A \boldsymbol{\mu} + \text{tr}(A\Sigma)$.

18. [Rao73a, p. 522] Let $\mathbf{y}$ be a $k \times 1$ random vector with expectation $\boldsymbol{\mu} = E(\mathbf{y})$ and covariance matrix $\Sigma = \text{var}(\mathbf{y})$, and let $[\boldsymbol{\mu} \mid \Sigma]$ denote the $k \times (k+1)$ partitioned matrix with $\boldsymbol{\mu}$ as its first column. Then $\mathbf{y} - \boldsymbol{\mu} \in \text{range}(\Sigma)$ and $\mathbf{y} \in \text{range}([\boldsymbol{\mu} \mid \Sigma])$, both with probability 1.
19. Let the $(k+q) \times 1$ random vector $\mathbf{x}$ have covariance matrix Σ . Consider the following partitioning:

$$\mathbf{x} = \begin{bmatrix} \mathbf{y} \\ \mathbf{z} \end{bmatrix}, \quad E(\mathbf{x}) = \begin{bmatrix} \boldsymbol{\mu} \\ \boldsymbol{\nu} \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \Sigma_{yy} & \Sigma_{yz} \\ \Sigma_{zy} & \Sigma_{zz} \end{bmatrix},$$

where $\mathbf{y}$ and $\mathbf{z}$ have k and q components, respectively. Then

- The variance $\text{var}(\mathbf{a}^T \mathbf{y} + \mathbf{b}^T \mathbf{z}) = \mathbf{a}^T \Sigma_{yy} \mathbf{a} + 2\mathbf{a}^T \Sigma_{yz} \mathbf{b} + \mathbf{b}^T \Sigma_{zz} \mathbf{b}$ for all nonrandom $\mathbf{a} \in \mathbb{R}^k$ and for all nonrandom $\mathbf{b} \in \mathbb{R}^q$.
- The covariance matrix $\text{var}(A\mathbf{y} + B\mathbf{z}) = A\Sigma_{yy}A^T + A\Sigma_{yz}B^T + B\Sigma_{zy}A^T + B\Sigma_{zz}B^T$ for all $A \in \mathbb{R}^{n \times k}$ and all $B \in \mathbb{R}^{n \times q}$.
- [PS05b, pp. 187–188] For any $A \in \mathbb{R}^{q \times k}$ the covariance matrix

$$\text{var}(\mathbf{z} - A\mathbf{y}) \succeq \text{var}(\mathbf{z} - \Sigma_{zy}\Sigma_{yy}^{-}\mathbf{y})$$

with respect to the partial semidefinite ordering, and the partial covariance matrix

$$\text{var}(\mathbf{z} - \Sigma_{zy}\Sigma_{yy}^{-}\mathbf{y}) = \Sigma_{zz} - \Sigma_{zy}\Sigma_{yy}^{-}\Sigma_{yz} = \Sigma_{zz \cdot y} = \Sigma / \Sigma_{yy},$$

the generalized Schur complement of Σ_{yy} in Σ .

- Let $q = k$. Then the covariance matrix $\text{var}(\mathbf{y} + \mathbf{z}) = \text{var}(\mathbf{y}) + \text{var}(\mathbf{z})$ if and only if $\text{cov}(\mathbf{y}, \mathbf{z}) = -\text{cov}(\mathbf{z}, \mathbf{y})$, i.e., the cross-covariance matrix $\text{cov}(\mathbf{y}, \mathbf{z})$ is skew-symmetric; the condition that $\text{cov}(\mathbf{y}, \mathbf{z}) = \mathbf{0}$ is sufficient, but not necessary (unless $k = 1$).
- The vector $\Sigma_{zy}\Sigma_{yy}^{-}\mathbf{y}$ is not necessarily invariant with respect to the choice of generalized inverse Σ_{yy}^{-} , but its covariance matrix $\text{var}(\Sigma_{zy}\Sigma_{yy}^{-}\mathbf{y}) = \Sigma_{zy}\Sigma_{yy}^{-}\Sigma_{yz}$ is invariant (and, hence, unique).

Examples:

1. Let the 4×1 random vector $\mathbf{x}$ have covariance matrix Σ . Consider the following partitioning:

$$\mathbf{x} = \begin{bmatrix} \mathbf{y} \\ \mathbf{z} \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 1 & 0 & a & b \\ 0 & 1 & c & d \\ a & c & 1 & 0 \\ b & d & 0 & 1 \end{bmatrix},$$

where $\mathbf{y}$ and $\mathbf{z}$ each have 2 components. Then $\text{var}(\mathbf{y} + \mathbf{z}) = \text{var}(\mathbf{y}) + \text{var}(\mathbf{z})$ if and only if $a = d = 0$ and $c = -b$, with $b^2 \leq 1$.

2. Let the $k \times 1$ random vector $\mathbf{y} = [Y_1, \dots, Y_k]^T$ follow a multinomial distribution with parameters n and $\mathbf{p} = [p_1, \dots, p_k]^T$, with $p_1 + \dots + p_k = 1$ and $p_1 > 0, \dots, p_k > 0$, and let the $k \times k$ matrix $A = [a_{ij}]$. Then the expectation $E(\mathbf{y}^T A \mathbf{y}) = n(n-1)\mathbf{p}^T A \mathbf{p} + n \sum_{i=1}^k a_{ii} p_i$.
3. Let the 3×1 random vector $\mathbf{y} = [Y_1, Y_2, Y_3]^T$ follow a trinomial distribution with parameters n and p_1, p_2, p_3 , with $p_1 + p_2 + p_3 = 1$ and $p_1 > 0, p_2 > 0, p_3 > 0$, and let the 3×1 vector $\mathbf{p} = [p_1, p_2, p_3]^T$. Then:
- The expectation $E(\mathbf{y}) = n[p_1, p_2, p_3]^T$ and the covariance matrix

$$\Psi_3 = \text{var}(\mathbf{y}) = n \begin{bmatrix} p_1(1-p_1) & -p_1 p_2 & -p_1 p_3 \\ -p_1 p_2 & p_2(1-p_2) & -p_2 p_3 \\ -p_1 p_3 & -p_2 p_3 & p_3(1-p_3) \end{bmatrix},$$

which has rank equal to 2 since Ψ_3 is singular and the determinant of the top left-hand corner of Ψ_3 equals $n^2 p_1 p_2 p_3 > 0$.

- The partial covariance matrix of Y_1 and Y_2 adjusting for Y_3 is the Schur complement $\Psi_3 / (p_3(1 - p_3)) = nS$, say, where

$$S = \begin{bmatrix} p_1(1 - p_1) & -p_1 p_2 \\ -p_1 p_2 & p_2(1 - p_2) \end{bmatrix} - \begin{bmatrix} -p_1 p_3 \\ -p_2 p_3 \end{bmatrix} \frac{1}{p_3(1 - p_3)} \begin{bmatrix} -p_1 p_3 & -p_2 p_3 \end{bmatrix} = \frac{p_1 p_2}{p_1 + p_2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix},$$

which has rank equal to 1, and so $\text{rank}(\Psi_3) = 2$.

- When $p_1 = p_2 = p_3 = 1/3$, then the covariance matrix $\Psi_3 = (n/3)C_3$ and the partial covariance matrix of Y_1 and Y_2 adjusting for Y_3 is $(n/3)C_2$; here, C_h is the $h \times h$ centering matrix, $h = 2, 3$.
4. [PS05, p. 183] If the 3×3 symmetric matrix

$$R_3 = \begin{bmatrix} 1 & r_{12} & r_{13} \\ r_{12} & 1 & r_{23} \\ r_{13} & r_{23} & 1 \end{bmatrix} = \begin{bmatrix} R_2 & \mathbf{r}_2 \\ \mathbf{r}_2^T & 1 \end{bmatrix}$$

is a correlation matrix, then $r_{ij}^2 \leq 1$ for all $1 \leq i < j \leq 3$. But not all symmetric matrices with diagonal elements all equal to 1 and all off-diagonal elements r_{ij} such that $r_{ij}^2 \leq 1$ are correlation matrices. For example, consider R_3 with $r_{13}^2 \leq 1$ and $r_{23}^2 \leq 1$. Then R_3 is a correlation matrix if and only if

$$r_{13}r_{23} - \sqrt{(1 - r_{13}^2)(1 - r_{23}^2)} \leq r_{12} \leq r_{13}r_{23} + \sqrt{(1 - r_{13}^2)(1 - r_{23}^2)}.$$

When $r_{13} = 0$ and $r_{12} = r_{23} = r$, say, then this condition becomes $r^2 \leq 1/2$ and so the matrix

$$\begin{bmatrix} 1 & 0.8 & 0 \\ 0.8 & 1 & 0.8 \\ 0 & 0.8 & 1 \end{bmatrix}$$

is not a correlation matrix.

When $r_{12}^2 \leq 1$, then the matrix R_3 is a correlation matrix if and only if any one of the following conditions holds:

- $\det(R_3) = 1 - r_{12}^2 - r_{13}^2 - r_{23}^2 + 2r_{12}r_{13}r_{23} \geq 0$.
- (i) $\mathbf{r}_2 \in \text{range}(R_2)$ and (ii) $1 \geq \mathbf{r}_2^T R_2^- \mathbf{r}_2$ for some and, hence, for every generalized inverse R_2^- .

5. Let the random vector $\mathbf{x}$ be 2×1 and write

$$\mathbf{x} = \begin{bmatrix} Y \\ Z \end{bmatrix}, \quad E(\mathbf{x}) = \begin{bmatrix} \mu \\ \nu \end{bmatrix}, \quad \text{var}(\mathbf{x}) = \Sigma = \begin{bmatrix} \sigma_y^2 & \sigma_{yz} \\ \sigma_{yz} & \sigma_z^2 \end{bmatrix},$$

with $\sigma_y^2 > 0$. Then the residual vector $\mathbf{e}_{\mathbf{z} \cdot \mathbf{y}} = \mathbf{z} - \boldsymbol{\nu} - \Sigma_{zy} \Sigma_{yy}^- (\mathbf{y} - \boldsymbol{\mu})$ of the random vector $\mathbf{z}$ from its regression on $\mathbf{y}$ becomes the scalar residual

$$e_{Z \cdot Y} = Z - \nu - \frac{\sigma_{yz}}{\sigma_y^2} (Y - \mu)$$

of the random variable Z from its regression on Y . The matrix of partial covariances of the random vector $\mathbf{z}$ after adjusting for $\mathbf{y}$ becomes the single partial variance

$$\sigma_{z \cdot y}^2 = \sigma_z^2 - \frac{\sigma_{yz}^2}{\sigma_y^2} = \sigma_z^2(1 - \rho_{yz}^2)$$

of the random variable Z after adjusting for the random variable Y ; here, the correlation coefficient $\rho_{yz} = \sigma_{yz}/(\sigma_y\sigma_z)$.

52.4 Linear Statistical Models: Basic Definitions and Facts

Notation:

In this section, the uppercase, light-face italic letter X is reserved for the nonrandom $n \times p$ model matrix and V is reserved for an $n \times n$ covariance matrix. The uppercase, light-face italic letter H is reserved for the (symmetric idempotent) $n \times n$ hat matrix $X(X^T X)^{-1} X^T$ and $M = I - H$ is reserved for the (symmetric idempotent) $n \times n$ residual matrix. The lowercase, bold-face roman letter $\mathbf{y}$ is reserved for an observable $n \times 1$ random vector and $\mathbf{x}$ is reserved for a column of the $n \times p$ model matrix X .

Definitions:

The **general linear model** (or **Gauss–Markov model** or **Gauß–Markov model**) is the model

$$\mathcal{M} = \{\mathbf{y}, X\boldsymbol{\beta}, \sigma^2 V\}$$

defined by the equation $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where $E(\mathbf{y}) = X\boldsymbol{\beta}$, $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, $\text{var}(\mathbf{y}) = \text{var}(\boldsymbol{\varepsilon}) = \sigma^2 V$. The vector $\mathbf{y}$ is an $n \times 1$ observable random vector, $\boldsymbol{\varepsilon}$ is an $n \times 1$ unobservable random error vector, X is a known $n \times p$ **model matrix** (or **design matrix**, particularly when its entries are $-1, 0$, or $+1$), $\boldsymbol{\beta}$ is a $p \times 1$ vector of unknown parameters, V is a known $n \times n$ positive semidefinite matrix, and σ^2 is an unknown positive constant. The realization of the $n \times 1$ observable random vector $\mathbf{y}$ will also be denoted by $\mathbf{y}$.

The classical theory of linear statistical models covers the **full-rank model**, where X has full column rank and V is positive definite. In the full-rank model, the **ordinary least squares estimator**

$$\text{OLSE}(\boldsymbol{\beta}) = \hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{y} = X^\dagger \mathbf{y}$$

and the **generalized least squares estimator** (or **Aitken estimator**)

$$\text{GLSE}(\boldsymbol{\beta}) = \tilde{\boldsymbol{\beta}} = (X^T V^{-1} X)^{-1} X^T V^{-1} \mathbf{y},$$

where $X^\dagger$ denotes the Moore–Penrose inverse of X .

When either X or V is (or both X and V are) rank deficient, then it is usually assumed that $\text{rank}(X) < \text{rank}(V)$. The model $\mathcal{M} = \{\mathbf{y}, X\boldsymbol{\beta}, \sigma^2 V\}$ is called a **weakly singular model** (or **Zyskind–Martin model**) whenever $\text{range}(X) \subseteq \text{range}(V)$, and then $\text{rank}(X) < \text{rank}(V)$, and is **consistent** if the realization $\mathbf{y}$ satisfies $\mathbf{y} \in \text{range}([X \mid V])$.

Let $\hat{\boldsymbol{\beta}}$ be any vector minimizing $\|\mathbf{y} - X\boldsymbol{\beta}\|^2 = (\mathbf{y} - X\boldsymbol{\beta})^T (\mathbf{y} - X\boldsymbol{\beta})$. Then $\hat{\mathbf{y}} = X\hat{\boldsymbol{\beta}} = \text{OLSE}(X\boldsymbol{\beta}) =$ the **ordinary least squares estimator** (OLSE) of $X\boldsymbol{\beta}$. When $\text{rank}(X) < p$, then $\hat{\boldsymbol{\beta}}$ is an **ordinary least squares solution** to $\min_{\boldsymbol{\beta}} (\mathbf{y} - X\boldsymbol{\beta})^T (\mathbf{y} - X\boldsymbol{\beta})$. Moreover, $\hat{\boldsymbol{\beta}}$ is any solution to the **normal equations** $X^T X\hat{\boldsymbol{\beta}} = X^T \mathbf{y}$. The vector of OLS **residuals** is $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - X\hat{\boldsymbol{\beta}}$ and the **residual sum of squares** $\text{SSE} = \mathbf{e}^T \mathbf{e} = (\mathbf{y} - \hat{\mathbf{y}})^T (\mathbf{y} - \hat{\mathbf{y}})$.

The **coefficient of determination** (or **coefficient of multiple determination** or **squared multiple correlation**) $R^2 = 1 - (\text{SSE} / \mathbf{y}^T C_n \mathbf{y})$ identifies the proportion of variance explained in a **multiple linear regression** where the model matrix $X = [\mathbf{1}_n \mid \mathbf{x}_{[1]} \mid \cdots \mid \mathbf{x}_{[p-1]}]$ with $p - 1$ **regressor vectors** (or **regressors**) $\mathbf{x}_{[1]}, \dots, \mathbf{x}_{[p-1]}$ each $n \times 1$. In **simple linear regression** $p = 2$ and the model matrix $X = [\mathbf{1}_n \mid \mathbf{x}]$

with the single regressor vector $\mathbf{x}$. The **sample correlation coefficient** $r = \mathbf{x}^T C_n \mathbf{y} / \sqrt{\mathbf{x}^T C_n \mathbf{x} \cdot \mathbf{y}^T C_n \mathbf{y}}$, where it is usually assumed that $\mathbf{x}$ is an $n \times 1$ nonrandom vector (such as a regressor vector) and $\mathbf{y}$ is a realization of the $n \times 1$ random vector $\mathbf{y}$.

Let the matrix $A \in \mathbb{R}^{k \times n}$ and let the matrix $K \in \mathbb{R}^{k \times p}$. Then the linear estimator $A\mathbf{y}$ is a **linear unbiased estimator** (LUE) of $K\beta$ if $E(A\mathbf{y}) = K\beta$ for all $\beta \in \mathbb{R}^p$. Let the matrix $B \in \mathbb{R}^{k \times n}$. Then the LUE $B\mathbf{y}$ of $K\beta$ is the **best linear unbiased estimator** (BLUE) of $K\beta$ if it has the smallest covariance matrix (in the positive semidefinite ordering) in that $\text{var}(A\mathbf{y}) \succeq \text{var}(B\mathbf{y})$ for all LUEs $A\mathbf{y}$ of $K\beta$.

The **hat matrix** $H = X(X^T X)^- X^T$ associated with the model matrix X is so named since $\hat{\mathbf{y}} = H\mathbf{y}$. The **residual matrix** $M = I - H$ and vector of OLS residuals is $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - H\mathbf{y} = M\mathbf{y}$. Let the nonrandom vector $\mathbf{a} \in \mathbb{R}^n$. Then the linear estimator $\mathbf{a}^T \mathbf{y}$, which is unbiased for 0, i.e., $E(\mathbf{a}^T \mathbf{y}) = 0$, is a **linear zero function**.

The **Watson efficiency** ϕ under the full-rank model $\mathcal{M} = \{\mathbf{y}, X\beta, \sigma^2 V\}$, with the $n \times p$ model matrix X having full column rank equal to $p < n$ and with the $n \times n$ covariance matrix V positive definite, measures the relative efficiency of the OLSE(β) = $\hat{\beta}$ vs. the BLUE(β) = $\tilde{\beta}$ and is defined by the ratio of the corresponding generalized variances:

$$\phi = \frac{\det[\text{var}(\tilde{\beta})]}{\det[\text{var}(\hat{\beta})]} = \frac{\det^2(X^T X)}{\det(X^T V X) \cdot \det(X^T V^{-1} X)}.$$

The **Bloomfield–Watson efficiency** ψ under the general linear model $\mathcal{M} = \{\mathbf{y}, X\beta, \sigma^2 V\}$ with no rank assumptions measures the relative efficiency of the OLSE($X\beta$) = $X\hat{\beta}$ vs. the BLUE(β) = $\tilde{\beta}$ and is defined by: $\psi = \frac{1}{2} \|HV - VH\|^2 = \|HVM\|^2$, where the norm $\|A\| = \text{tr}^{1/2}(A^T A)$ is defined for any $k \times q$ matrix A .

The $n \times n$ covariance matrix $(1 - \rho)I_n + \rho \mathbf{1}_n \mathbf{1}_n^T = (1 - \rho)I_n + \rho J_n$ has **intraclass correlation structure** (or **equicorrelation structure**) and is the **intraclass correlation matrix** (or the **equicorrelation matrix**). The parameter ρ is the **intraclass correlation** (or **intraclass correlation coefficient**).

Facts:

The following facts, except for those with a specific reference, can be found in [Gro04], [PS89], or [SJ03, §4.1–4.3]. Throughout this set of facts, X denotes the $n \times p$ nonrandom model matrix.

1. The hat matrix $H = X(X^T X)^- X^T$ associated with the model matrix X is invariant (unique) with respect to choice of generalized inverse $(X^T X)^-$ and is a symmetric idempotent matrix: $H = H^T = H^2$, and $\text{rank}(H) = \text{tr}(H) = \text{rank}(X)$. Moreover, the hat matrix H is the orthogonal projector onto $\text{range}(X)$.
2. If the $p \times p$ matrix Q is nonsingular, then the hat matrix associated with the model matrix XQ equals the hat matrix associated with the model matrix X .
3. The residual sum of squares $SSE = \mathbf{y}^T M \mathbf{y} = (\mathbf{y} - \hat{\mathbf{y}})^T (\mathbf{y} - \hat{\mathbf{y}}) = \mathbf{y}^T \mathbf{y} - \mathbf{y}^T X \hat{\beta}$, where M is the residual matrix and $\hat{\beta} = \text{OLSE}(\beta)$.
4. In simple linear regression the coefficient of determination $R^2 = r^2$, the square of the sample correlation coefficient. In multiple linear regression with model matrix $X = [\mathbf{1}_n | X_0] = [\mathbf{1}_n | \mathbf{x}_{[1]} | \cdots | \mathbf{x}_{[p-1]}]$ and $(p - 1) \times 1$ nonrandom vector $\mathbf{a} \in \mathbb{R}^p$,

$$R^2 = \max_{\mathbf{a}} r_{\mathbf{a}}^2 = \max_{\mathbf{a}} \frac{(\mathbf{a}^T X_0^T C_n \mathbf{y})^2}{\mathbf{a}^T X_0^T C_n X_0 \mathbf{a} \cdot \mathbf{y}^T C_n \mathbf{y}},$$

the square of the sample correlation coefficient $r_{\mathbf{a}}$ between the variables whose observed values are in vectors $\mathbf{y}$ and $X_0 \mathbf{a}$.

5. The vector $X\hat{\beta}$ is invariant (unique) with respect to the choice of least squares solution $\hat{\beta}$, but $\hat{\beta}$ is unique if and only if X has full column rank equal to $p \leq n$, and then $\hat{\beta} = \text{OLSE}(\beta) = (X^T X)^-1 X^T \mathbf{y} = X^\dagger \mathbf{y}$, where $X^\dagger$ is the Moore–Penrose inverse of X . The covariance matrix $\text{var}(\hat{\beta}) = \sigma^2 (X^T X)^-1 X^T V X (X^T X)^-1$.

6. The Watson efficiency ϕ is always positive, and $\phi \leq 1$ with equality if and only if $\text{OLSE}(\beta) = \text{BLUE}(\beta)$.
7. [DLL02, p. 477], [Gus97, p. 67] *Bloomfield–Watson–Knott Inequality*. The Watson efficiency

$$\phi = \frac{\det^2(X^T X)}{\det(X^T V X) \cdot \det(X^T V^{-1} X)} \geq \prod_{i=1}^m \frac{4\lambda_i \lambda_{n-i+1}}{(\lambda_i + \lambda_{n-i+1})^2},$$

for all $n \times p$ model matrices X with full column rank p . Here $m = \min(p, n-p)$ and $\lambda_1 \geq \cdots \geq \lambda_n$ denote the necessarily positive eigenvalues of the $n \times n$ positive definite covariance matrix V . The ratios $4\lambda_i \lambda_{n-i+1}/(\lambda_i + \lambda_{n-i+1})^2$ in the lower bound for the Watson efficiency are the squared *antieigenvalues* of the covariance matrix V .

8. [DLL02, p. 454] Let $p = 1$ and set the $n \times 1$ model matrix $X = \mathbf{x}$. Then the Bloomfield–Watson–Knott Inequality is the *Kantorovich Inequality* (or *Frucht–Kantorovich Inequality*):

$$\frac{(\mathbf{x}^T \mathbf{x})^2}{\mathbf{x}^T V \mathbf{x} \cdot \mathbf{x}^T V^{-1} \mathbf{x}} \geq \frac{4\lambda_1 \lambda_n}{(\lambda_1 + \lambda_n)^2},$$

where λ_1 and λ_n are, respectively, the largest and smallest eigenvalues of the $n \times n$ positive definite covariance matrix V .

9. The Bloomfield–Watson efficiency

$$\begin{aligned} \psi &= \frac{1}{2} \|HV - VH\|^2 = \|HVM\|^2 = \text{tr}(HVMVH) = \text{tr}(HVMV) \\ &= \text{tr}(HV^2 - HVHV) = \text{tr}(HV^2) - \text{tr}((HV)^2) \geq 0, \end{aligned}$$

with equality if and only if $\text{OLSE}(\beta) = \text{BLUE}(\beta)$ if and only if the Watson efficiency $\phi = 1$.

10. [DLL02, p. 473] *The Bloomfield–Watson Trace Inequality*. Let A be a nonrandom symmetric $n \times n$ matrix, not necessarily positive semidefinite. Then for all the nonrandom matrices $U \in \mathbb{R}^{n \times p}$ that satisfy $U^T U = I_p$:

$$\text{tr}(U^T A^2 U) - \text{tr}((U^T A U)^2) \leq \frac{1}{4} \sum_{i=1}^{\min(p, n-p)} (\alpha_i - \alpha_{n-i+1})^2,$$

where $\alpha_1 \geq \cdots \geq \alpha_n$ denote the eigenvalues of the $n \times n$ matrix A .

11. The Bloomfield–Watson efficiency

$$\psi = \text{tr}(HV^2) - \text{tr}((HV)^2) \leq \frac{1}{4} \sum_{i=1}^{\min(p, n-p)} (\lambda_i - \lambda_{n-i+1})^2,$$

for all $n \times n$ hat matrices H with rank p (and so for all $n \times p$ model matrices X with full column rank p). Here, $\lambda_1 \geq \cdots \geq \lambda_n$ denote the necessarily positive eigenvalues of the $n \times n$ positive definite covariance matrix V .

12. The $n \times n$ intraclass correlation matrix $R_{\text{ic}} = (1 - \rho)I_n - \rho \mathbf{1}_n \mathbf{1}_n^T$ has eigenvalues $1 - \rho$ with multiplicity $n - 1$ and $1 + \rho(n - 1)$ with multiplicity 1, and so R_{ic} is singular if and only if $\rho = -1/(n - 1)$ or $\rho = 1$.
13. The intraclass correlation coefficient ρ is such that $-1/(n - 1) \leq \rho \leq 1$ and the $n \times n$ intraclass correlation matrix is positive definite if and only if $-1/(n - 1) < \rho < 1$.
14. The inverse of the $n \times n$ positive definite intraclass correlation matrix

$$((1 - \rho)I_n - \rho \mathbf{1}_n \mathbf{1}_n^T)^{-1} = \frac{1}{1 - \rho} \left(I_n - \frac{\rho}{1 + \rho(n - 1)} \mathbf{1}_n \mathbf{1}_n^T \right).$$

15. *Gauss–Markov Theorem* (or *Gauß–Markov Theorem*). In the full-rank model $\{\mathbf{y}, X\boldsymbol{\beta}, \sigma^2 V\}$, the generalized least squares estimator $\tilde{\boldsymbol{\beta}} = \text{GLSE}(\boldsymbol{\beta}) = (X^T V^{-1} X)^{-1} X^T V^{-1} \mathbf{y} = \text{BLUE}(\boldsymbol{\beta})$. In the full-rank model $\{\mathbf{y}, X\boldsymbol{\beta}, \sigma^2 I\}$, the ordinary least-squares estimator $\text{OLSE}(\boldsymbol{\beta}) = \hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{y} = X^+ \mathbf{y} = \text{BLUE}(\boldsymbol{\beta})$.
16. In the model $\{\mathbf{y}, X\boldsymbol{\beta}, \sigma^2 V\}$, where V is positive definite, but with X possibly with less than full column rank, the

$$\text{BLUE}(X\boldsymbol{\beta}) = X(X^T V^{-1} X)^{-} X^T V^{-1} \mathbf{y}.$$

17. [Sea97, §5.4] Let the matrix $K \in \mathbb{R}^{k \times p}$. Then $K\boldsymbol{\beta}$ is estimable $\iff \exists$ matrix $A \in \mathbb{R}^{n \times k}$: $K^T = X^T A \iff \text{range}(K^T) \subseteq \text{range}(X^T) \iff K\tilde{\boldsymbol{\beta}}$ is invariant for any choice of $\tilde{\boldsymbol{\beta}} = (X^T X)^{-} X^T \mathbf{y}$.
18. [Rao73b, p. 282] Consider the general linear model $\{\mathbf{y}, X\boldsymbol{\beta}, \sigma^2 V\}$, where X and V need not be of full rank. Let the matrix $G \in \mathbb{R}^{n \times n}$. Then $G\mathbf{y} = \text{BLUE}(X\boldsymbol{\beta}) \iff G[X | VM] = [X | \mathbf{0}]$, where the residual matrix $M = I - H$. Let the matrix $A \in \mathbb{R}^{k \times n}$ and the matrix $K \in \mathbb{R}^{k \times p}$. Then the corresponding condition for $A\mathbf{y}$ to be the BLUE of an estimable parametric function $K\boldsymbol{\beta}$ is $A[X | VM] = [K | \mathbf{0}]$.
19. Let G_1 and G_2 both be $n \times n$. If $G_1 \mathbf{y}$ and $G_2 \mathbf{y}$ are two BLUEs of $X\boldsymbol{\beta}$ under the model $\{\mathbf{y}, X\boldsymbol{\beta}, \sigma^2 V\}$, then $G_1 \mathbf{y} = G_2 \mathbf{y}$ for all $\mathbf{y} \in \text{range}([X | V])$. The matrix G yielding the BLUE is unique if and only if $\text{range}([X | V]) = \mathbb{R}^n$.
20. Every linear zero function can be written as $\mathbf{b}^T M \mathbf{y}$ for some nonrandom $\mathbf{b} \in \mathbb{R}^n$. Let the matrix $G \in \mathbb{R}^{n \times n}$. Then an unbiased estimator $G\mathbf{y} = \text{BLUE}(X\boldsymbol{\beta})$ if and only if $G\mathbf{y}$ is uncorrelated with every linear zero function.
21. [Rao71] Let the matrix $A \in \mathbb{R}^{n \times n}$. Then the linear estimator $A\mathbf{y} = \text{BLUE}(X\boldsymbol{\beta})$ under the model $\{\mathbf{y}, X\boldsymbol{\beta}, \sigma^2 V\}$ if and only if there exists a matrix Λ so that A is a solution to *Pandora's box*

$$\begin{bmatrix} V & X \\ X^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} A^T \\ \Lambda \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ X^T \end{bmatrix}.$$

22. [Rao71] Let the $(n + p) \times (n + p)$ matrix B be defined as any generalized inverse:

$$B = \begin{bmatrix} V & X \\ X^T & \mathbf{0} \end{bmatrix}^- = \begin{bmatrix} B_1 & B_2 \\ B_3 & -B_4 \end{bmatrix}.$$

Let $\mathbf{k}^T \boldsymbol{\beta}$ be estimable; then the $\text{BLUE}(\mathbf{k}^T \boldsymbol{\beta}) = \mathbf{k}^T \tilde{\boldsymbol{\beta}} = \mathbf{k}^T B_2^T \mathbf{y} = \mathbf{k}^T B_3 \mathbf{y}$, the variance $\text{var}(\mathbf{k}^T \tilde{\boldsymbol{\beta}}) = \sigma^2 \mathbf{k}^T B_4 \mathbf{k}$, and the quadratic form $\mathbf{y}^T B_1 \mathbf{y} / f$ is an unbiased estimator of σ^2 with $f = \text{rank}([V | X]) - \text{rank}(X)$.

23. [PS89] In the model $\{\mathbf{y}, X\boldsymbol{\beta}, \sigma^2 V\}$ with no rank assumptions, the $\text{OLSE}(X\boldsymbol{\beta}) = \text{BLUE}(X\boldsymbol{\beta})$ if and only if any one of the following equivalent conditions holds:

- $HV = VH$.
- $HV = HVB$.
- $HVM = \mathbf{0}$.
- $X^T VL = \mathbf{0}$, where the $n \times l$ matrix L has $\text{range}(L) = \text{range}(M)$.
- $\text{range}(VX) \subseteq \text{range}(X)$.
- $\text{range}(VX) = \text{range}(X) \cap \text{range}(V)$.
- $HVH \leq V$, i.e., $V - HVH$ is positive semidefinite.
- $\text{rank}(V - HVH) = \text{rank}(V) - \text{rank}(HVH)$.
- $\text{rank}(V - HVB) = \text{rank}(V) - \text{rank}(VB)$.
- $\text{range}(X)$ has a basis consisting of r eigenvectors of V , where $r = \text{rank}(X)$.
- V can be expressed as $V = \alpha I + XAX^T + LBL^T$, where $\alpha \in \mathbb{R}$, $\text{range}(L) = \text{range}(M)$, and the $p \times p$ matrices A and B are symmetric, and such that V is positive semidefinite.

More conditions can be obtained by replacing V with its Moore–Penrose inverse $V^\dagger$ and the hat matrix H with the residual matrix $M = I - H$.

24. Suppose that the positive definite covariance matrix V has h distinct eigenvalues: $\lambda_{\{1\}} > \lambda_{\{2\}} > \cdots > \lambda_{\{h\}} > 0$ with multiplicities $m_1, \dots, m_h$, $\sum_{i=1}^h m_i = n$, and with associated orthonormalized sets of eigenvectors $U_{\{1\}}, \dots, U_{\{h\}}$, respectively, $n \times m_1, \dots, n \times m_h$. Then $\text{OLSE}(X\beta) = \text{BLUE}(X\beta)$ if and only if any one of the following equivalent conditions holds:
- $\text{rank}(U_{\{1\}}^T X) + \cdots + \text{rank}(U_{\{h\}}^T X) = \text{rank}(X)$.
 - $U_{\{i\}}^T H U_{\{i\}} = (U_{\{i\}}^T H U_{\{i\}})^2$ for all $i = 1, \dots, h$.
 - $U_{\{i\}}^T H U_{\{j\}} = \mathbf{0}$ for all $i \neq j$; $i, j = 1, \dots, h$.
25. [Rao73b] Let the $p \times p$ matrix U be such that the $n \times n$ matrix $W = V + XU X^T$ has $\text{range}(W) = \text{range}([X \mid V])$. Then the $\text{BLUE}(X\beta) = X(X^T W^- X)^- X^T W^- \mathbf{y}$.
26. When V is nonsingular, the $n \times n$ matrix G such that $G\mathbf{y}$ is the BLUE of $X\beta$ is unique, but when V is singular this may not be so. However, the numerical value of $\text{BLUE}(X\beta)$ is unique with probability 1.
27. [SJ03, §7.4] The residual vector associated with the $\text{BLUE}(X\beta)$ is

$$\tilde{\mathbf{e}} = \mathbf{y} - X\tilde{\beta} = VM(MVM)^- M\mathbf{y} = M\mathbf{y} + HVM(MVM)^- M\mathbf{y},$$

which is invariant (unique) with respect to choice of generalized inverse $(MVM)^-$. The weighted sum of squares of BLUE residuals, which is needed when estimating σ^2 , can be written as $(\mathbf{y} - X\tilde{\beta})^T V^- (\mathbf{y} - X\tilde{\beta}) = \tilde{\mathbf{e}}^T V^- \tilde{\mathbf{e}} = \mathbf{y}^T M(MVM)^- M\mathbf{y}$.

Examples:

1. Let $n = 3$ and $p = 2$ with the model matrix $X = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 1 & -1 \end{bmatrix}$. Then X has full column rank equal to 2, the matrix $X^T X$ is nonsingular, and the hat matrix is

$$H = X(X^T X)^- X^T = X(X^T X)^{-1} X^T = \frac{1}{6} \begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}$$

with $\text{rank}(H) = \text{tr}(H) = 2$. The $\text{OLSE}(\beta)$ is

$$\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y} = \begin{bmatrix} \frac{1}{3}(y_1 + y_2 + y_3) \\ \frac{1}{2}(y_1 - y_3) \end{bmatrix},$$

where $\mathbf{y} = [y_1, y_2, y_3]^T$. The vector of OLS residuals is

$$M\mathbf{y} = \frac{1}{6} \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \frac{1}{6}(y_1 - 2y_2 + y_3) \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

with residual sum of squares $SSE = (y_1 - 2y_2 + y_3)^2/6$.

Now let the variance $\sigma^2 = 1$ and let the covariance matrix

$$V = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \delta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

with $\delta > 0$. Then V is positive definite and

$$\text{BLUE}(\beta) = \text{GLSE}(\beta) = \tilde{\beta} = \begin{bmatrix} \frac{1}{2\delta+1}(\delta y_1 + y_2 + \delta y_3) \\ \frac{1}{2}(y_1 - y_3) \end{bmatrix},$$

while the covariance matrices are

$$\text{var}(\tilde{\beta}) = \begin{bmatrix} \frac{\delta}{2\delta+1} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}, \quad \text{var}(\hat{\beta}) = \begin{bmatrix} \frac{\delta+2}{9} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}.$$

Hence, the Watson efficiency

$$\phi = \frac{9\delta}{(2\delta+1)(\delta+2)} \leq 1$$

with equality if and only if $\delta = 1$. As $\delta \rightarrow 0$ or $\delta \rightarrow \infty$, the Watson efficiency $\phi \rightarrow 0$.

Since the eigenvalues of V here are 1 (multiplicity 2) and δ (multiplicity 1), we find that the lower bound for ϕ in the Bloomfield–Watson–Knott Inequality [here, $m = \min(p, n - p) = \min(2, 1) = 1$] is equal to $4\delta/(1+\delta)^2$, and it is easy to show that this is less than $9\delta/((2\delta+1)(\delta+2))$ (unless $\delta = 1$, and then both these ratios are equal to 1).

The Bloomfield–Watson efficiency

$$\psi = \frac{2}{9}(1 - \delta)^2 \geq 0$$

with equality if and only if $\delta = 1$. As $\delta \rightarrow 0$ the Bloomfield–Watson efficiency $\psi \rightarrow 2/9$, and as $\delta \rightarrow \infty$ the Bloomfield–Watson efficiency $\psi \rightarrow 0$. We note that when $\delta = 0$, then Σ is singular, but ψ is well-defined and equal to $2/9$. We find that the upper bound for ψ in the Bloomfield–Watson Trace Inequality is $(1 - \delta)^2/4$ and certainly $(1 - \delta)^2/4 \geq 2(1 - \delta)^2/9$ with equality if and only if $\delta = 1$.

2. Let $n = 3$ and $p = 1$ with the model matrix $X = \mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and with $\beta = \beta$, a scalar. The hat matrix

$$H = X(X^T X)^{-1} X^T = \frac{1}{\mathbf{x}^T \mathbf{x}} \mathbf{x} \mathbf{x}^T = \frac{1}{14} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

and so

$$\hat{\beta} = \text{OLSE}(\beta) = \frac{\mathbf{x}^T \mathbf{y}}{\mathbf{x}^T \mathbf{x}} = \frac{y_1 + 2y_2 + 3y_3}{14},$$

where the realization $\mathbf{y} = [y_1, y_2, y_3]^T$.

Let the covariance matrix have intraclass correlation structure:

$$V = \begin{bmatrix} 1 & \rho & \rho \\ \rho & 1 & \rho \\ \rho & \rho & 1 \end{bmatrix}$$

with $-1/2 < \rho < 1$. Then V is positive definite and its inverse

$$V^{-1} = \frac{1}{(1 - \rho)(1 + 2\rho)} \begin{bmatrix} 1 + \rho & -\rho & -\rho \\ -\rho & 1 + \rho & -\rho \\ -\rho & -\rho & 1 + \rho \end{bmatrix}.$$

And so

$$\tilde{\beta} = \text{BLUE}(\beta) = \text{GLSE}(\beta) = \frac{\mathbf{x}^T V^{-1} \mathbf{y}}{\mathbf{x}^T V^{-1} \mathbf{x}} = \frac{(1-4\rho)y_1 + 2(1-\rho)y_2 + 3y_3}{6(1-\rho)}.$$

The variances are (with $\sigma^2 = 1$):

$$\text{var}(\tilde{\beta}) = \frac{1}{\mathbf{x}^T V^{-1} \mathbf{x}} = \frac{(1-\rho)(1+2\rho)}{2(7-4\rho)}; \quad \text{var}(\hat{\beta}) = \frac{\mathbf{x}^T V \mathbf{x}}{(\mathbf{x}^T \mathbf{x})^2} = \frac{7+11\rho}{98},$$

and so the Watson efficiency

$$\phi = \frac{\text{var}(\tilde{\beta})}{\text{var}(\hat{\beta})} = \frac{49(1-\rho)(1+2\rho)}{(7-4\rho)(7+11\rho)} \rightarrow 0$$

as $\rho \rightarrow -1/2$ or as $\rho \rightarrow 1$. As $\rho \rightarrow 0$ the Watson efficiency $\phi \rightarrow 1$.

Since the eigenvalues of V here are $1-\rho$ (multiplicity 2) and $1+2\rho$ (multiplicity 1), we find that the lower bound for ϕ in the Bloomfield–Watson–Knott Inequality (which is now the Kantorovich Inequality) is $4(1-\rho)(1+2\rho)/(2+\rho)^2$ and it is easy to show that this is less than $\phi = 49(1-\rho)(1+2\rho)/((7-4\rho)(7+11\rho))$ (unless $\rho = 0$ and then both these ratios are equal to 1).

The Bloomfield–Watson efficiency $\psi = 54\rho^2/49$, which is well defined for all ρ ; when $\rho = -1/2$, then $\psi = 27/98$ and when $\rho = 1$, then $\psi = 54/49$. We find that the upper bound for ψ in the Bloomfield–Watson Trace Inequality is $9\rho^2/4$ and certainly $9\rho^2/4 \geq \psi = 54\rho^2/49$, with equality if and only if $\rho = 0$.

3. [DLL02, p. 475] Let A be a nonrandom symmetric $n \times n$ matrix, not necessarily positive semidefinite, and let the nonrandom vector $\mathbf{u} = [u_1, \dots, u_n]^T$ be such that $\mathbf{u}^T \mathbf{u} = 1$. Then from the Bloomfield–Watson Trace Inequality with $U = \mathbf{u}$, we obtain the special case

$$\mathbf{u}^T A^2 \mathbf{u} - (\mathbf{u}^T A \mathbf{u})^2 \leq \frac{1}{4}(\alpha_1 - \alpha_n)^2,$$

where α_1 and α_n are, respectively, the largest and smallest eigenvalues of A . Now let $a_1, \dots, a_n$ denote n nonrandom scalars, not necessarily all positive, and let $\bar{a} = \sum_{i=1}^n a_i/n$. Then the Popoviciu–Nair Inequality

$$\frac{1}{n} \sum_{i=1}^n (a_i - \bar{a})^2 \leq \frac{1}{4}(a_{\max} - a_{\min})^2$$

follows directly from the special case above of the Bloomfield–Watson Trace Inequality with $\mathbf{u} = \mathbf{1}/\sqrt{n}$ and $A = \text{diag}\{a_i\}$.

4. When the covariance matrix V has intraclass correlation structure, then the $\text{OLSE}(X\beta) = \text{BLUE}(X\beta)$ if and only if $X^T \mathbf{1}_n = \mathbf{0}$ or $X\mathbf{f} = \mathbf{1}_n$ for some nonrandom $p \times 1$ vector $\mathbf{f}$, and so $\text{OLSE}(X\beta) = \text{BLUE}(X\beta)$ when the columns of X are centered or when $\mathbf{1}_n \in \text{range}(X)$ as in Example 1

above, where $X = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 1 & -1 \end{bmatrix}$ with first column equal to $\mathbf{1}_3$.

5. Let the $n \times p$ model matrix $X = [\mathbf{1} \mid X_0]$, where $\mathbf{1}$ is the $n \times 1$ column vector with every entry equal to 1 and X_0 is $n \times (p-1)$. Then the hat matrix associated with X ,

$$H = \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^T + C_n X_0 (X_0^T C_n X_0)^- X_0^T C_n$$

coincides with the hat matrix associated with $X_c = [\mathbf{1}_n \mid C_n X_0]$, since $X_c = XQ$, with the $p \times p$ matrix $Q = \begin{bmatrix} 1 & -\mathbf{q}^T \\ \mathbf{0} & I_{p-1} \end{bmatrix}$ for some $\mathbf{q}$.

When $p = 2$, we may write $X_0 = \mathbf{x}$, an $n \times 1$ vector, and if X now has rank equal to 2, then $\mathbf{x}^T C_n \mathbf{x} > 0$ and the hat matrix

$$H = \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^T + \frac{1}{\mathbf{x}^T C_n \mathbf{x}} C_n \mathbf{x} \mathbf{x}^T C_n = \frac{1}{n} J_n + \frac{1}{\mathbf{x}^T C_n \mathbf{x}} C_n \mathbf{x} \mathbf{x}^T C_n.$$

And so the quadratic forms

$$\mathbf{y}^T H \mathbf{y} = n \bar{y}^2 + \frac{(\mathbf{x}^T C_n \mathbf{y})^2}{\mathbf{x}^T C_n \mathbf{x}}, \quad \mathbf{y}^T M \mathbf{y} = \mathbf{y}^T C_n \mathbf{y} + \frac{(\mathbf{x}^T C_n \mathbf{y})^2}{\mathbf{x}^T C_n \mathbf{x}},$$

where $\bar{y} = \sum_{i=1}^n y_i / n = \mathbf{1}_n^T \mathbf{y} / n$.

Acknowledgments

We are very grateful to Ka Lok Chu, Anne Greenbaum, Leslie Hogben, Jarkko Isotalo, Augustyn Markiewicz, George A. F. Seber, Evelyn Matheson Styan, Götz Trenkler, and Kimmo Vehkalahti for their help. This research was supported in part by the Natural Sciences and Engineering Research Council of Canada.

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53

Multivariate Statistical Analysis

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Vectors and matrices arise naturally in the analysis of statistical data and we have seen this in Chapter 52. For example, suppose we take a random sample of n females and measure their heights x_i ($i = 1, 2, \dots, n$), giving us the vector $\mathbf{x} = [x_1, x_2, \dots, x_n]^T$. We can treat $\mathbf{x}$ as simply a random vector, which can give rise to different values depending on the sample chosen, or as a vector of observed values (the data) taken on by the random vector from which we calculate a statistic like a sample mean (average). We use the former approach when we want to make inferences about the female population. In this case, we find that the value of a given x_i will depend on how it varies across the population, and this is described by its statistical “distribution,” which is defined by a univariate function of one variable called a probability density function (pdf). For example, x_i may follow the well-known normal distribution, which seems to apply to many naturally occurring measurements. This distribution has a probability density function totally characterized by two (unknown) parameters, the population mean μ and the population variance σ^2 , where σ is the standard deviation; we write $x_i \sim N_1(\mu, \sigma^2)$, where “ $\sim$ ” means “distributed as.” (Throughout this chapter σ will always refer to a standard deviation and not to a singular value.) When the sample is random, the choice of any female does not affect the choice of any other, so that technically we say that the x_i are statistically independent and they all have the same distribution.

Three important aspects of statistical inference are: (1) estimating μ from the data and deriving the distributional properties of the estimate from the assumed underlying pdf; (2) finding a so-called confidence interval for μ , which is an interval containing the true value of μ with a given probability, e.g., 0.95 or, when typically expressed as a percentage, 95%; and (3) testing a hypothesis (called the null hypothesis H_0) about whether μ is equal to some predetermined value μ_0 , say. Symbolically, we write $H_0 : \mu = \mu_0$. To test H_0 , we need to have a test statistic and be able to derive its distribution when H_0 is true and when

it is false, called the noncentral distribution in the latter case. Noncentral distributions are usually quite complex, but they have a role in determining how good a test is in detecting departures from the null hypothesis.

In multivariate analysis we are interested in measuring not just a single characteristic or variable on each female, but we may also want to record weight, income, blood pressure, and so on, so that each x_i gets replaced by a d -dimensional vector $\mathbf{x}_i$ of d variables. Then $\mathbf{x}$ from the sample of recorded heights gets replaced by a matrix X with rows $\mathbf{x}_i^T$. In this case the probability density function for $\mathbf{x}_i$ is said to be multivariate and the population mean is now a vector $\boldsymbol{\mu}$, say. We can look at X as either a matrix of numbers, or a random matrix, which we use for carrying out statistical inferences. In doing this, the univariate normal distribution extends naturally to the multivariate normal distribution, which has a pdf totally characterized by its mean $\boldsymbol{\mu}$ and its covariance matrix $\text{var}(\mathbf{x}) = \Sigma$. One can then derive other distributions based on various functions of X , which can then be used for inference.

The vectors $\mathbf{x}_i$ can be regarded as a cluster of points in $\mathbb{R}^d$. Such a set of points can also come from sampling a mixed population of men and women so that there would be two overlapping clusters. Alternatively, the population of females could be a mixture of races, each with its own characteristics, leading to several overlapping clusters. In order to study such clusters, we need techniques to somehow reduce the dimension of the data (hopefully, to two dimensions) in some optimal fashion, but still retain as much of the original variation as possible. We will introduce just three dimension-reducing techniques below. The first method, called principal component analysis, is used for a single cluster. The second method, called discriminant coordinates, is used for several clusters and the reduction in dimension is carried out to maximize group separation. The third method, called canonical variates, is also used for a single cluster, but utilizes the internal variation within the vectors. Looking at how one variable within a vector depends on the other variables lends itself to a range of other techniques, such as multivariate linear regression where we might compare the internal relationships in one cluster with those in others. For example, comparing how blood pressure depends on other variables for both males and females.

We finally introduce just one other topic called metric multidimensional scaling. Instead of having observations on n people or objects, we simply have measures of similarity or dissimilarity between each pair of objects. The challenge then is to try and represent these objects as a cluster in a low-dimensional space so that the inter-point Euclidean distances will closely reproduce the dissimilarity measures. Once we have the cluster we can then examine it to try and uncover any underlying structure or clustering of the objects.

It is hoped that these few applications will demonstrate the richness of the subject and its interplay between statistics and matrices. In this chapter we assume some of the basic statistical ideas defined in Chapter 52. However, there has to be a change in notation as we now need to continually distinguish between random and nonrandom vectors and matrices.

53.1 Data Matrix

Notation:

The latter part of the alphabet from u to z , upper or lower case, together with Q we reserve for random quantities, and the remainder of the alphabet for nonrandom quantities. All quantities in this section are real, though in practice some of the theory can be extended to complex quantities, as, for example, in the theory of time series.

Definitions:

If $\mathbf{w} = [\mathbf{y}^T, \mathbf{z}^T]^T$ is a random vector, then $\mathbf{y}$ and $\mathbf{z}$ are said to be **uncorrelated** if and only if their cross-covariance $\text{cov}(\mathbf{y}, \mathbf{z}) = \mathbf{0}$. We say that $\mathbf{y}$ and $\mathbf{z}$ are **statistically independent** if and only if the probability density function (pdf) of $\mathbf{w}$ is the product of the pdfs of $\mathbf{y}$ and $\mathbf{z}$.

Let $X = [x_{ij}] = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_n^T \end{bmatrix} = [\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(d)}]$ be an $n \times d$ matrix of random variables with rows $\mathbf{x}_i^T$ such

that all $\mathbf{x}_i$ have the same covariance matrix Σ and are uncorrelated, i.e., $\text{cov}(\mathbf{x}_r, \mathbf{x}_s) = \delta_{rs} \Sigma$, where $\delta_{rs} = 1$ for $r = s$ and $\delta_{rs} = 0$ for $r \neq s$. We call a matrix with the above properties a **data matrix**. As mentioned in the introduction, the $\mathbf{x}_i$ often constitute a random sample, which we now formally define. A **random sample** of vectors $\mathbf{x}_i$ ($i = 1, 2, \dots, n$) of size n consists of n random vectors that are **independently and identically distributed (i.i.d.)**, that is, they are statistically independent of each other and have the same pdf. When this is the case, we see from the data matrix above that the n elements of $\mathbf{x}^{(j)}$ form a random sample from the j th characteristic or variable, being just the j th elements in each $\mathbf{x}_i$. However, $\mathbf{x}^{(j)}$ and $\mathbf{x}^{(k)}$ ($j \neq k$) can be correlated.

The **moment generating function** or m.g.f. of any random vector $\mathbf{x}$ is defined to be $M_{\mathbf{x}}(\mathbf{t}) = E(e^{\mathbf{t}^T \mathbf{x}})$. It can be used for finding such things as the mean and covariance matrix, or the distribution of related random variables.

Facts:

1. For a random variable x , $M_x(t) = E(e^{xt}) = \sum_{r=0}^{\infty} E(x^r) \frac{t^r}{r!}$. The coefficient of $\frac{t^r}{r!}$ in the power series expansion of $M_x(t)$ gives us $E(x^r)$ from which we can get $\mu = E(x)$ and $\text{var}(x) = E(x^2) - \mu^2$.
2. Statistical independence implies zero covariance but not generally vice versa except for one notable exception, the multivariate normal distribution (see next section).

53.2 Multivariate Normal Distribution

Definitions:

Let $\mathbf{x} = [x_i] = [x_1, x_2, \dots, x_d]^T$ be a $d \times 1$ random vector with mean $\boldsymbol{\mu}$ and positive definite covariance matrix Σ . Then $\mathbf{x}$ is said to follow a (**nonsingular**) **multivariate normal distribution** when its pdf is

$$f(\mathbf{x} : \boldsymbol{\mu}, \Sigma) = (2\pi)^{-d/2} (\det \Sigma)^{-1/2} \exp\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1/2}(\mathbf{x} - \boldsymbol{\mu})\},$$

where $-\infty < x_j < \infty$, $j = 1, 2, \dots, d$. We write $\mathbf{x} \sim N_d(\boldsymbol{\mu}, \Sigma)$. When $d = 1$ we have the univariate normal distribution. When Σ is singular, the pdf of $\mathbf{x}$ does not exist but it can be defined via a transformation $\mathbf{x} = A\mathbf{y}$ where $\mathbf{y}$ does have a nonsingular distribution. This is mentioned further below.

The **noncentral χ^2 -distribution** with p degrees of freedom and noncentrality parameter δ , denoted by $\chi_{p,\delta}^2$, is the distribution of $\mathbf{x}^T \mathbf{x}$ when $\mathbf{x} \sim N_p(\boldsymbol{\mu}, I_p)$ and $\delta = \boldsymbol{\mu}^T \boldsymbol{\mu}$. The distribution is said to be **central** whenever $\delta = 0$, and we denote it by χ_p^2 .

The **noncentral F-distribution** with m, n degrees of freedom and noncentrality parameter δ , denoted by $F(m, n; \delta)$, is the distribution of the so-called **F-ratio** $F = \frac{u/m}{v/n}$, where $u \sim \chi_{m,\delta}^2$, $v \sim \chi_n^2$, and u and v are independent. The distribution is **central** whenever $\delta = 0$ and we then write $F \sim F(m, n)$.

Facts:

All the following facts except those with a specific reference can be found in [And03, Ch. 2], [Rao73, Sec. 8a], or [SL03, Ch. 1].

We can extend our definition to include the singular multivariate normal distribution using one of the first two facts below as a definition, each of which includes both the nonsingular and singular cases. When we write $\mathbf{x} \sim N_d(\boldsymbol{\mu}, \Sigma)$, we include both possibilities unless otherwise stated.

1. Assuming for the trivial case that $y \sim N_1(b, 0)$ means $y = b$ with probability 1, then the random $d \times 1$ vector $\mathbf{x}$ follows a multivariate normal distribution if and only if $y = \mathbf{a}^T \mathbf{x}$ is univariate normal for all $d \times 1$ vectors $\mathbf{a}$.
2. A random $d \times 1$ vector $\mathbf{x}$ with mean $\boldsymbol{\mu}$ and covariance matrix Σ follows a multivariate normal distribution of rank $m \leq d$ when it has the same distribution as $A\mathbf{z} + \boldsymbol{\mu}$, where A is any $d \times m$ matrix satisfying $\Sigma = AA^T$ and $\mathbf{z} \sim N_m(\mathbf{0}, I_m)$.
3. Given $\mathbf{x} \sim N_d(\boldsymbol{\mu}, \Sigma)$, then the m.g.f. of $\mathbf{x}$ is $M_{\mathbf{x}}(\mathbf{t}) = e^{\mathbf{t}^T \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^T \Sigma \mathbf{t}}$. When Σ is positive definite, this m.g.f. uniquely determines the nonsingular multivariate normal distribution.
4. Given $\mathbf{x} \sim N_d(\boldsymbol{\mu}, \Sigma)$ and A $m \times d$, then $A\mathbf{x} \sim N_m(A\boldsymbol{\mu}, A\Sigma A^T)$. The distribution is nonsingular if and only if $A\Sigma$ has full row rank m ($m \leq d$), which is assured when Σ is positive definite and A has rank m .
5. Any subset of the components of a multivariate normal random variable is multivariate normal.
6. If the cross-covariance matrix of any two vectors which contain disjoint subsets of $\mathbf{x}$ is zero, then the two vectors are statistically independent.
7. If the cross-covariance matrix $\text{cov}(A\mathbf{x}, B\mathbf{x}) = \mathbf{0}$, then $A\mathbf{x}$ and $B\mathbf{x}$ are statistically independent.
8. If Σ is positive definite, then $(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \sim \chi_d^2$, the central χ^2 -distribution with d degrees of freedom.
9. Let $\mathbf{x}$ be an $n \times 1$ random vector with $E(\mathbf{x}) = \boldsymbol{\mu}$ and $\text{var}(\mathbf{x}) = \Sigma$, and having any distribution. If A is any symmetric $n \times n$ matrix, then $E[(\mathbf{x} - \mathbf{a})^T A (\mathbf{x} - \mathbf{a})] = \text{tr}(A\Sigma) + (\boldsymbol{\mu} - \mathbf{a})^T A (\boldsymbol{\mu} - \mathbf{a})$.
10. Let $\mathbf{x} = [x_i]$ be an $n \times 1$ random vector with $E(x_i) = \theta_i$ and $\mu_r = E(x_i - \theta_i)^r$; $r = 2, 3, 4$. If A is any $n \times n$ symmetric matrix and $\mathbf{a}$ is the column vector of the diagonal elements of A , then

$$\text{var}(\mathbf{x}^T A \mathbf{x}) = (\mu_4 - 3\mu_2^2) \mathbf{a}^T \mathbf{a} + 2\mu_2^2 \text{tr} A^2 + 4\mu_2 \boldsymbol{\theta}^T A^2 \boldsymbol{\theta} + 4\mu_3 \boldsymbol{\theta}^T A \mathbf{a}.$$

Examples:

1. If $x \sim N_1(0, \sigma^2)$, then $M_x(t) = e^{\frac{1}{2}\sigma^2 t^2}$ from Fact 3 above. Since $\mu = 0$, $\mu_r = E(x^r)$ can be found by expanding $M_x(t)$ and finding the coefficient of $t^r/r!$. For example, $\mu_2 = \sigma^2$, $\mu_3 = 0$, and $\mu_4 = 3\mu_2^2$.
2. If the x_i ($i = 1, 2, \dots, n$) are i.i.d. as $N_1(0, \sigma^2)$, then by either multiplying pdfs together for independent random variables or using m.g.f.s we find that $\mathbf{x} \sim N_n(\mathbf{0}, \sigma^2 I_n)$. Substituting the results from the previous example into Fact 10 we get $\text{var}(\mathbf{x}^T A \mathbf{x}) = 2\sigma^4 \text{tr} A^2$.

53.3 Inference for the Multivariate Normal

Definitions:

Given a random sample $\{\mathbf{x}_i\}$ that are i.i.d., from the nonsingular multivariate normal distribution, the **likelihood function** is defined to be the joint pdf of the sample expressed as a function of the unknown parameters, namely,

$$L(\mathbf{u}, \Sigma) = \prod_{i=1}^n f(\mathbf{x}_i; \boldsymbol{\mu}, \Sigma).$$

The parameter estimates that maximize this function are called the **maximum likelihood estimates**. The **sample mean** of the sample is defined to be $\bar{\mathbf{x}} = \sum_{i=1}^n \mathbf{x}_i / n$, and is not to be confused with the complex conjugate.

Facts:

1. The maximum likelihood estimates of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are, respectively, $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$ and $\hat{\boldsymbol{\Sigma}} = Q/n = \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T/n$. Here, “ $\hat{}$ ” denotes “estimate of” in statistics.
2. $Q = X^T C X$, where X is the data matrix previously defined and the centering matrix $C = I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^T$ is symmetric and idempotent. Also, $CX = [\mathbf{x}_1 - \bar{\mathbf{x}}, \dots, \mathbf{x}_n - \bar{\mathbf{x}}]^T = \tilde{X}$, say, and $Q = \tilde{X}^T \tilde{X}$.

53.4 Principal Component Analysis

Definitions:

Let $\mathbf{x}$ be a random d -dimensional vector with mean $\boldsymbol{\mu}$ and positive definite covariance matrix $\boldsymbol{\Sigma}$. Let $T = [\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_d]$ be an orthogonal matrix such that $T^T \boldsymbol{\Sigma} T = \Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_d)$, where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d > 0$ are the ordered eigenvalues of $\boldsymbol{\Sigma}$. The sum $\text{tr} \boldsymbol{\Sigma}$ is sometimes called the **total variance**. If $\mathbf{y} = [y_j] = T^T(\mathbf{x} - \boldsymbol{\mu})$, then $y_j = \mathbf{t}_j^T(\mathbf{x} - \boldsymbol{\mu})$ ($j = 1, 2, \dots, d$) is called the j th **population principal component** of $\mathbf{x}$, and $z_j = \lambda_j^{-1/2} y_j$ is called the j th **standardized population principal component**.

In practice, $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are unknown and have to be estimated from a sample $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$. Assuming that the underlying distribution is multivariate normal we can use the estimates of the previous section, $\bar{\mathbf{x}}$ and $\hat{\boldsymbol{\Sigma}} = \tilde{X}^T \tilde{X}/n$. Carrying out a similar factorization on $\hat{\boldsymbol{\Sigma}}$ as we did for $\boldsymbol{\Sigma}$, we obtain the eigenvalues $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \geq \hat{\lambda}_d > 0$ and an orthogonal matrix $\hat{T} = [\hat{\mathbf{t}}_1, \hat{\mathbf{t}}_2, \dots, \hat{\mathbf{t}}_d]$ of corresponding eigenvectors. For each observation $\mathbf{x}_i$, we can define a vector $\hat{\mathbf{y}}_i = \hat{T}^T(\mathbf{x}_i - \bar{\mathbf{x}})$ of **sample principal components** or **estimated principal components** yielding $\hat{Y}^T = [\hat{\mathbf{y}}_1, \hat{\mathbf{y}}_2, \dots, \hat{\mathbf{y}}_n] = \hat{T}^T \tilde{X}^T$.

Although the above method is mainly used descriptively, asymptotic inference for large n can be carried out on the assumption that the underlying distribution is normal.

Facts:

The following facts can be found in [Chr01, Sec. 3.1–3.3], [Rao73, Sec. 8g2], or in [Seb04, Sec. 5.2].

1. As $\text{var}(\mathbf{y}) = \Lambda$, which is diagonal, the y_j are uncorrelated and $\text{var}(y_j) = \lambda_j$.
2. $\sum_{j=1}^d \text{var}(y_j) = \sum_{j=1}^d \text{var}(x_j) = \text{tr} \boldsymbol{\Sigma}$. We can use $\lambda_j/\text{tr} \boldsymbol{\Sigma}$ to measure the relative magnitude of λ_j . If the λ_i ($i = k+1, \dots, d$) are relatively small so that the corresponding y_i are “small” (with zero means and small variances), then $\mathbf{y}_{(k)} = [y_1, y_2, \dots, y_k]^T$ can be regarded as a k -dimensional approximation for $\mathbf{y}$. Thus, $\mathbf{y}_{(k)}$ can be used as a “proxy” for $\mathbf{x}$ in terms of explaining a major part of the total variance.
3. Let $T_{(k)} = [\mathbf{t}_1, \dots, \mathbf{t}_k]$. Then:
 - $\max_{\mathbf{a}^T \mathbf{a} = 1} \text{var}(\mathbf{a}^T \mathbf{x}) = \text{var}(\mathbf{t}_1^T \mathbf{x}) = \text{var}(\mathbf{t}_1^T(\mathbf{x} - \boldsymbol{\mu})) = \text{var}(y_1) = \lambda_1$, so that y_1 is the normalized linear combination of the elements of $\mathbf{x} - \boldsymbol{\mu}$ with maximum variance λ_1 .
 - $\max_{\mathbf{a}^T \mathbf{a} = 1, T_{(k-1)}^T \mathbf{a} = \mathbf{0}} \text{var}(\mathbf{a}^T \mathbf{x}) = \text{var}(\mathbf{t}_k^T \mathbf{x}) = \text{var}(y_k) = \lambda_k$, so that $\mathbf{t}_k^T(\mathbf{x} - \boldsymbol{\mu})$ is the normalized linear combination of the elements $\mathbf{x} - \boldsymbol{\mu}$ uncorrelated with $y_1, y_2, \dots, y_{k-1}$, with maximum variance λ_k .
4. $\hat{y}_{ij} = \hat{\mathbf{t}}_j^T(\mathbf{x}_i - \bar{\mathbf{x}})$, the score of the i th individual on the j th sample principal component, is related to the orthogonal projection of $\mathbf{x}_i - \bar{\mathbf{x}}$ onto $\text{range}(\hat{\mathbf{t}}_j)$, namely $P_{\hat{\mathbf{t}}_j}(\mathbf{x}_i - \bar{\mathbf{x}}) = \hat{\mathbf{t}}_j \hat{\mathbf{t}}_j^T(\mathbf{x}_i - \bar{\mathbf{x}}) = \hat{y}_{ij} \hat{\mathbf{t}}_j$.

Examples:

1. Suppose we assume that each $\mathbf{x}_i$ is just an observed vector, i.e., a constant. Let $\mathbf{v}$ be a discrete random vector taking the values $\mathbf{x}_i$ ($i = 1, 2, \dots, n$) with probability $\frac{1}{n}$. Then $E(\mathbf{v}) = \sum_{i=1}^n \mathbf{x}_i P(\mathbf{v} = \mathbf{x}_i) =$

$\sum_{i=1}^n \mathbf{x}_i / n = \bar{\mathbf{x}}$ and, similarly, $\text{var}(\mathbf{v}) = \hat{\Sigma}$. This means that the sample principal components for $\mathbf{x}$ are the population components for $\mathbf{v}$ so that all the optimal properties of population principal components hold correspondingly for the sample principal components.

- Entomologists are interested in the number of distinct taxa present in a population of winged aphids as they are difficult to identify. Forty aphids were trapped and 19 variables on each aphid were measured. A principal component analysis was carried out and the first two components accounted for 85% of the estimate of the total variance giving an effective reduction from 19 to 2 dimensions. The 2-dimensional $\hat{\mathbf{y}}_i$ were plotted showing the presence of four groups. Also, the $\hat{\mathbf{t}}_r$ ($r = 1, 2$) suggested which two linear combinations of the 19 measurements gave the best discrimination. As the aphids came from slightly different populations, the original assumption that the sample is from a homogeneous population is not quite true, but it does provide a starting place for further study, e.g., use discriminant coordinates described in the next section with four groups.

53.5 Discriminant Coordinates

Definitions:

Suppose we have n d -dimensional observations of which n_i belong to group i from the i th underlying population ($i = 1, 2, \dots, g$; $n = \sum_{i=1}^g n_i$). Let $\mathbf{x}_{ij}$ be the j th observation in group i , and define

$$\bar{\mathbf{x}}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \mathbf{x}_{ij} \quad \text{and} \quad \bar{\mathbf{x}}_{..} = \frac{1}{n} \sum_{i=1}^g \sum_{j=1}^{n_i} \mathbf{x}_{ij}.$$

Let $W_g = \sum_{i=1}^g \sum_{j=1}^{n_i} (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)(\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)^T$, the **within-groups matrix**, and let $W_b = \sum_{i=1}^g n_i (\bar{\mathbf{x}}_i - \bar{\mathbf{x}}_{..})(\bar{\mathbf{x}}_i - \bar{\mathbf{x}}_{..})^T$, the **between-groups matrix**. Since W_g and W_b are positive definite with probability 1, the eigenvalues of $W_g^{-1} W_b$, which are the same as those of $W_g^{-1/2} W_b W_g^{-1/2}$, are positive and distinct with probability 1, say $\lambda_1 > \lambda_2 > \dots > \lambda_d > 0$.

Let $W_g^{-1} W_b \mathbf{c}_r = \lambda_r \mathbf{c}_r$ be suitably scaled eigenvectors and define $C^T = [\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_k]$ ($k \leq d$). If we define $\mathbf{y}_{ij} = C \mathbf{x}_{ij}$, then the k elements of $\mathbf{z}_{ij}$ are called the first k **discriminant coordinates** (or **canonical variates**). These coordinates are determined so as to emphasize group separation, but with decreasing effectiveness so that k has to be determined. The coordinates can be computed using an appropriate transformation combined with a principal component analysis. Typically the $\mathbf{c}_i$ are scaled so that $C S C^T = I_r$, where $S = W_g / (n - g)$.

Examples:

- Trivariate measurements $\mathbf{x}_{ij}$ were taken on the skulls of six collections (four subspecies with three collections apparently from the same species) of anteaters. Using six groups of data from six underlying populations in the above theory, we found that $[\lambda_1, \lambda_2, \lambda_3] = [2.400, 0.905, 0.052]$. Since λ_3 was small, we chose $k = 2$. We could, therefore, transform the 3-dimensional observations $\mathbf{x}_{ij}$ into the 2-dimensional $\mathbf{y}_{ij} = C \mathbf{x}_{ij}$ and still account for most of the variation in the data. The 2-dimensional reductions were then plotted to help us look for any patterns. As an aid to our search, it was possible to draw a circle with center the reduced mean of each group so that the circle contained any reduced random observation from its population with probability 0.95. The center of the circle locates the center of gravity of the group and the boundary of the circle shows where the bulk of the observations are expected to lie. The closeness of these circles indicated how “close” the groups and subspecies were. We found that three of the circles almost overlapped completely, confirming a common underlying species, and the remaining circles had little overlap, suggesting the presence of four species.

53.6 Canonical Correlations and Variates

Definitions:

Let $\mathbf{z} = \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}$ denote a d -dimensional random vector with mean $\boldsymbol{\mu}$ and positive definite covariance matrix Σ . Let $\mathbf{x}$ and $\mathbf{y}$ have dimensions d_1 and $d_2 = d - d_1$, respectively, and consider the partition

$$\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix},$$

where Σ_{ii} is $d_i \times d_i$ and $\Sigma_{21} = \Sigma_{12}^T$ has rank r . Let ρ_1^2 be the maximum value of the squared correlation between arbitrary linear combinations $\boldsymbol{\alpha}^T \mathbf{x}$ and $\boldsymbol{\beta}^T \mathbf{y}$, and let $\boldsymbol{\alpha} = \mathbf{a}_1$ and $\boldsymbol{\beta} = \mathbf{b}_1$ be the corresponding maximizing values of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$. Then the positive square root $\sqrt{\rho_1^2}$ is called the **first (population) canonical correlation** between $\mathbf{x}$ and $\mathbf{y}$, and $u_1 = \mathbf{a}_1^T \mathbf{x}$ and $v_1 = \mathbf{b}_1^T \mathbf{y}$ are called the **first (population) canonical variables**.

Let ρ_2^2 be the maximum value of the squared correlation between $\boldsymbol{\alpha}^T \mathbf{x}$ and $\boldsymbol{\beta}^T \mathbf{y}$, where $\boldsymbol{\alpha}^T \mathbf{x}$ is uncorrelated with $\mathbf{a}_1^T \mathbf{x}$ and $\boldsymbol{\beta}^T \mathbf{y}$ is uncorrelated with $\mathbf{b}_1^T \mathbf{y}$, and let $u_2 = \mathbf{a}_2^T \mathbf{x}$ and $v_2 = \mathbf{b}_2^T \mathbf{y}$ be the maximizing values. Then the positive square root $\sqrt{\rho_2^2}$ is called the **second canonical correlation**, and u_2 and v_2 are called the **second canonical variables**. Continuing in this manner, we obtain r pairs of canonical variables $\mathbf{u} = [u_1, u_2, \dots, u_r]^T$ and $\mathbf{v} = [v_1, v_2, \dots, v_r]^T$. We can then regard $\mathbf{u}$ and $\mathbf{v}$ as lower-dimensional “representations” of $\mathbf{x}$ and $\mathbf{y}$.

Facts:

- [Seb04, Sec. 5.7] If $m = \text{rank} \Sigma_{12} \geq 1$ and Σ is positive definite, then $\Sigma_{11}^{-1} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$ has m positive eigenvalues $\rho_1^2 \geq \rho_2^2 \geq \dots \geq \rho_m^2$, say, and $\rho_1^2 < 1$. Moreover, $\Sigma_{11}^{-1} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$ and $\Sigma_{22}^{-1} \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}$ have the same (nonzero) eigenvalues. Let $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m$ and $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_m$ be the corresponding eigenvectors of $\Sigma_{11}^{-1} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$ and $\Sigma_{22}^{-1} \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}$, respectively. Suppose that $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are arbitrary vectors such that for $r \leq m$, $\boldsymbol{\alpha}^T \mathbf{x}$ is uncorrelated with each $\mathbf{a}_j^T \mathbf{x}$ ($j = 1, 2, \dots, r-1$), and $\boldsymbol{\beta}^T \mathbf{y}$ is uncorrelated with each $\mathbf{b}_j^T \mathbf{y}$ ($j = 1, 2, \dots, r-1$). Let $u_j = \mathbf{a}_j^T \mathbf{x}$ and $v_j = \mathbf{b}_j^T \mathbf{y}$, for $j = 1, 2, \dots, r$. Then:
 - The maximum squared correlation between $\boldsymbol{\alpha}^T \mathbf{x}$ and $\boldsymbol{\beta}^T \mathbf{y}$ is given by ρ_r^2 and it occurs when $\boldsymbol{\alpha} = \mathbf{a}_r$ and $\boldsymbol{\beta} = \mathbf{b}_r$.
 - $\text{cov}(u_j, u_k) = 0$ for $j \neq k$, and $\text{cov}(v_j, v_k) = 0$ for $j \neq k$.
 - The squared correlation between u_j and v_j is ρ_j^2 .
 - Since ρ_j^2 is independent of scale, we can scale $\mathbf{a}_j$ and $\mathbf{b}_j$ such that $\mathbf{a}_j^T \Sigma_{11} \mathbf{a}_j = 1$ and $\mathbf{b}_j^T \Sigma_{22} \mathbf{b}_j = 1$. The u_j and v_j then have unit variances.
 - If the $d_1 \times d_2$ matrix Σ_{12} has row full rank, and $d_1 < d_2$, then $m = d_1$. All the eigenvalues of $\Sigma_{11}^{-1} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$ are then positive, while $\Sigma_{22}^{-1} \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}$ has d_1 positive eigenvalues and $d_2 - d_1$ zero eigenvalues. The rank of Σ_{12} can vary as there may be constraints on Σ_{12} , such as $\Sigma_{12} = \mathbf{0}$ (rank 0) or $\Sigma_{12} = \mathbf{a} \mathbf{1}_{d_1}^T \mathbf{1}_{d_1}$ (rank 1).
- [Bri01, p. 370]. Suppose $\mathbf{x}$ and $\mathbf{y}$ have means $\boldsymbol{\mu}_x$ and $\boldsymbol{\mu}_y$, respectively. Let $\mathbf{u} = A(\mathbf{x} - \boldsymbol{\mu}_x)$ and $\mathbf{v} = B(\mathbf{y} - \boldsymbol{\mu}_y)$, where A and B are any matrices, each with r rows that are linearly independent, satisfying $A \Sigma_{11} A^T = I_r$ and $B \Sigma_{22} B^T = I_r$. Then $E[(\mathbf{u} - \mathbf{v})^T (\mathbf{u} - \mathbf{v})]$ is minimized when $\mathbf{u}$ and $\mathbf{v}$ are vectors of the canonical variables.
- [BS93], [SS80] When Σ is singular, then the (population) canonical correlations are the positive square roots of the eigenvalues of $\Sigma_{11}^{-} \Sigma_{12} \Sigma_{22}^{-} \Sigma_{21}$; any generalized inverses Σ_{11}^{-} and Σ_{22}^{-} may be chosen. There will be u canonical correlations equal to 1, where $u = \text{rank}(\Sigma_{11}) + \text{rank}(\Sigma_{22}) - \text{rank}(\Sigma)$.

53.7 Estimation of Correlations and Variates

Definitions:

Let $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_n$ be a random sample and let $\bar{\mathbf{z}} = \frac{1}{n} \sum_{i=1}^n \mathbf{z}_i = \begin{bmatrix} \bar{x} \\ \bar{y} \end{bmatrix}$ be the sample mean and $\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n (\mathbf{z}_i - \bar{\mathbf{z}})(\mathbf{z}_i - \bar{\mathbf{z}})^T$, where $\hat{\Sigma}$ is partitioned in the same way as Σ , namely

$$n\hat{\Sigma} = \begin{bmatrix} \tilde{X}^T \tilde{X} & \tilde{X}^T \tilde{Y} \\ \tilde{Y}^T \tilde{X} & \tilde{Y}^T \tilde{Y} \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{bmatrix},$$

say. Assuming that $d_1 \leq d_2$, let $r_1^2 > r_2^2 > \dots > r_{d_1}^2 > 0$ be the eigenvalues of $Q_{11}^{-1} Q_{12} Q_{22}^{-1} Q_{21}$, with corresponding eigenvectors $\hat{\mathbf{a}}_1, \hat{\mathbf{a}}_2, \dots, \hat{\mathbf{a}}_{d_1}$. We define $u_{ij} = \hat{\mathbf{a}}_j^T (\mathbf{x}_i - \bar{\mathbf{x}})$, the i th element of $\mathbf{u}_j = \tilde{X} \hat{\mathbf{a}}_j$, where $\hat{\mathbf{a}}_j$ is scaled so that $1 = n^{-1} \hat{\mathbf{a}}_j^T \tilde{X}^T \tilde{X} \hat{\mathbf{a}}_j = \sum_{i=1}^n u_{ij}^2 / n = \mathbf{u}_j^T \mathbf{u}_j / n$. Then $\sqrt{r_j^2}$ is the j th **sample canonical correlation**. Moreover, u_{ij} is the j th **sample canonical variable** of $\mathbf{x}_i$. In a similar fashion, we define $v_{ij} = \hat{\mathbf{b}}_j^T (\mathbf{y}_i - \bar{\mathbf{y}})$, the i th element of $\mathbf{v}_j = \tilde{Y} \hat{\mathbf{b}}_j$, to be the j th sample canonical variable of $\mathbf{y}_i$. Here, $\hat{\mathbf{b}}_1, \hat{\mathbf{b}}_2, \dots, \hat{\mathbf{b}}_{d_1}$ are the corresponding eigenvectors of $Q_{22}^{-1} Q_{21} Q_{11}^{-1} Q_{12}$, scaled so that $\mathbf{v}_j^T \mathbf{v}_j / n = 1$. The u_{ij} and v_{ij} are the **scores** of the i th observation on the j th canonical variables.

Facts:

1. When Σ is positive definite and $n - 1 \geq d$, then, with probability 1, $n\hat{\Sigma}$ is positive definite and $\text{rank } Q_{12} = d_1$.
2. The Canonical correlations $\sqrt{r_j^2}$ are all distinct with probability 1.
3. r_j^2 is the square of the sample correlation between the canonical variables whose values are in the vectors $\mathbf{u}_j$ and $\mathbf{v}_j$.

Examples:

1. Length and breadth head measurements were carried out on the first and second sons of 25 families to see what relationships existed. Here, $\mathbf{x}$ refers to the two measurements on the first son and $\mathbf{y}$ to the second. It was found that $r_1 = 0.7885$ and $r_2 = 0.0537$, indicating that just one pair of sample canonical variables (u_1, v_1) would give a reasonable reduction in dimension from 2 to 1. Plotting u_1 against v_1 using the 21 pairs (u_{i1}, v_{i1}) gave a reasonably linear plot. The first linear combinations (u_1, v_1) of the two measurements suggested that they could be interpreted as a measure of “girth” while the second (u_2, v_2) could be interpreted as “shape” measurements. We can conclude that there is a strong correlation between the head sizes of first and second brothers, but not between the head shapes.

53.8 Matrix Quadratic Forms

To carry out inference for multivariate normal distributions we now require further theory.

Definitions:

If X is an $n \times d$ data matrix and $A = [a_{ij}]$ is a symmetric $n \times n$ matrix, then the expression $X^T A X = \sum_{i=1}^n \sum_{j=1}^n a_{ij} \mathbf{x}_i \mathbf{x}_j^T$ is said to be a **matrix quadratic form**. We have already had one example, namely, $Q = X^T C X$ in section 53.3. When the $\{\mathbf{x}_i\}$ are a random sample from a distribution with mean $\boldsymbol{\mu}$ and covariance matrix Σ we define $S = Q/(n - 1)$ to be the **sample covariance matrix**. (Some authors define it as $\hat{\Sigma}$ in Section 53.3.)

If the $\mathbf{x}_i$ are i.i.d. as $N_d(\mathbf{0}, \Sigma)$, with $d \leq n$ and $\Sigma = [\sigma_{ij}]$ positive semidefinite, then $W = X^T X$ is said to have a **Wishart distribution** with n degrees of freedom and scale matrix Σ , and we write $W \sim W_d(n, \Sigma)$. If Σ is positive definite then the distribution is said to be **nonsingular**.

Suppose $\mathbf{y} \sim N_d(\mathbf{0}, \Sigma)$ independently of $W \sim W_d(m, \Sigma)$ and that both distributions are nonsingular. Then **Hotelling's T^2 distribution** is defined to be the distribution of $T^2 = m\mathbf{y}^T W^{-1} \mathbf{y}$ and we denote this distribution by $T^2 \sim T_{d,m}^2$.

Finally, in inference we are usually interested in **contrasts** $\mathbf{a}^T \boldsymbol{\mu}$, in the μ_i , where $\mathbf{a}$ is a vector whose entries sum to 0, i.e., $\mathbf{a}^T \mathbf{1}_d = 0$. Examples are $\mu_1 - \mu_2 = [1, -1, 0, \dots, 0] \boldsymbol{\mu}$ and $\mu_1 - \frac{1}{2}(\mu_2 + \mu_3)$.

Facts:

All facts, unless otherwise indicated, appear in [Seb04, Secs. 2.3–2.4, 3.3–3.4, 3.6.2].

1. $E(S) = \Sigma$, irrespective of the distribution of the $\mathbf{x}_i$.
2. [HS79] Let X be an $n \times d$ data matrix and let A and B be matrices of appropriate sizes. If vec is the usual “stacking” operator, then:
 - $\text{cov}[\text{vec}(AXB), \text{vec}(CXD)] = (B^T \Sigma D) \otimes (AC^T)$.
 - If $U = AXB$ and $V = CXD$, then U and V are pairwise uncorrelated, that is, $\text{cov}(u_{ij}, v_{rs}) = 0$ for all i, j, r , and s , if $AC^T = 0$ and/or $B^T \Sigma D = \mathbf{0}$.
3. Suppose $W = [w_{ij}]$ has a Wishart distribution $W_d(m, \Sigma)$, with Σ possibly singular. Then the following hold:
 - $E(W) = m\Sigma$.
 - Let A be a $q \times d$ matrix with $\text{rank } q \leq d$. Then $AWA^T \sim W_q(m, A\Sigma A^T)$. When Σ is positive definite, then this distribution is nonsingular.
 - When $\sigma_{jj} > 0$, then $w_{jj}/\sigma_{jj} \sim \chi_m^2$, ($j = 1, 2, \dots, d$). However, the w_{jj} ($j = 1, \dots, d$) are not statistically independent.
 - When Σ is positive definite, then $\det \Sigma > 0$ and $\det W / \det \Sigma$ is distributed as the product of d independent chi-square variables with respective degrees of freedom $m, m-1, \dots, m-d+1$.
4. (See [SK79, p. 97], [Sty89].) Let $W \sim W_d(m, \Sigma)$ with Σ possibly singular, and let A be an $n \times n$ symmetric matrix. Then:
 - $E(WAW) = m[\Sigma A^T \Sigma + \text{tr}(A\Sigma)\Sigma] + m^2 \Sigma A \Sigma$.
 - If $m > d+1$ and Σ is nonsingular, then

$$E(WAW^{-1}) = \frac{1}{m-d-1} [m\Sigma A \Sigma^{-1} - A^T - (\text{tr} A)I],$$

$$E(W^{-1}AW) = \frac{1}{m-d-1} [m\Sigma^{-1}A\Sigma - A^T - (\text{tr} A)I].$$

- If $m > d+3$ and Σ is nonsingular, then

$$E(W^{-1}AW^{-1}) = \frac{(m-d-2)\Sigma^{-1}A\Sigma^{-1} + \Sigma^{-1}A^T\Sigma^{-1} - (\text{tr} A\Sigma^{-1})\Sigma^{-1}}{(m-d)(m-d-1)(m-d-3)}.$$

5. Suppose that the rows $\mathbf{x}_i^T$ of X are uncorrelated with $\text{var}(\mathbf{x}_i) = \Sigma_i$ for $i = 1, 2, \dots, n$, and A is an $n \times n$ symmetric matrix. Then:
 - $E(X^T AX) = \sum_{i=1}^n a_{ii} \Sigma_i + E(X^T)AE(X)$.
 - If $\Sigma_i = \Sigma$ for all i , then $E(X^T AX) = (\text{tr} A)\Sigma + E(X^T)AE(X)$.
6. [Das71, Th. 5], [EP73, Th. 2.3] Suppose that the rows of X are independent and A is an $n \times n$ positive semidefinite matrix of rank $r \geq d$. If for each $\mathbf{x}_i$ and all $\mathbf{b}$ and c with $\mathbf{b} \neq \mathbf{0}$ the probability $\Pr(\mathbf{b}^T \mathbf{x}_i = c) = 0$, then $X^T AX$ is positive definite with probability 1.

7. Suppose that rows of X are i.i.d. as $N_d(\mathbf{0}, \Sigma)$, with Σ possibly singular, and let A and B be $n \times n$ symmetric matrices.
 - If A has rank $r \leq d$, then $X^T A X \sim W_d(r, \Sigma)$ if and only if $A = A^2$.
 - Suppose $X^T A X$ and $X^T B X$ have Wishart distributions. They are statistically independent if and only if $AB = \mathbf{0}$.
8. Referring to the definition of Hotelling's T^2 , $\frac{m-d+1}{md} T_{d,m}^2 \sim F(d, m-d+1)$, the F-distribution with d and $m-d+1$ degrees of freedom. When $\mathbf{y} \sim N_d(\boldsymbol{\theta}, \Sigma)$, then $F \sim F(d, m-d+1; \delta)$, the corresponding noncentral F-distribution with noncentrality parameter $\delta = \boldsymbol{\theta}^T \Sigma \boldsymbol{\theta}$.
9. Suppose that the rows of X are i.i.d. as nonsingular $N_d(\boldsymbol{\mu}, \Sigma)$. Then:
 - $\bar{\mathbf{x}} \sim N_d(\boldsymbol{\mu}, \Sigma/n)$.
 - $Q = (n-1)S \sim W_d(n-1, \Sigma)$.
 - $\bar{\mathbf{x}}$ and S are statistically independent.
 - $T^2 = n(\bar{\mathbf{x}} - \boldsymbol{\mu})^T S^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \sim T_{d,n-1}^2$. This statistic can be used for testing the null hypothesis $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$.
10. Given $H_0 : \boldsymbol{\mu} \in \mathcal{V}$, where $\mathcal{V}$ is a p -dimensional vector subspace of $\mathbb{R}^d$, then:
 - $T_{\min}^2 = \min_{\boldsymbol{\mu} \in \mathcal{V}} T^2 \sim T_{d-p,n-1}^2$.
 - If $H_0 : \boldsymbol{\mu} = K\boldsymbol{\beta}$, where K is a known $d \times p$ matrix of rank p and $\boldsymbol{\beta}$ is a vector of p unknown parameters, then $\mathcal{V} = \text{range}(K)$, and $T_{\min}^2 = n(\bar{\mathbf{x}}^T S^{-1} \bar{\mathbf{x}} - \bar{\mathbf{x}}^T S^{-1} K \boldsymbol{\beta}^*)$, where $\boldsymbol{\beta}^* = (K^T S^{-1} K)^{-1} K^T S^{-1} \bar{\mathbf{x}}$.
11. If $H_0 : A\boldsymbol{\mu} = \mathbf{0}$, where A is $(d-p) \times d$ of rank $d-p$, then $\mathcal{V} = \ker(A)$ (also called the null space) and $T_{\min}^2 = n(A\bar{\mathbf{x}})^T (ASA^T)^{-1} A\bar{\mathbf{x}}$.
 - Let A be a $q \times d$ matrix of rank $q \leq d$. Then the quadratic $n(A\bar{\mathbf{x}} - A\boldsymbol{\mu})^T (ASA^T)^{-1} (A\bar{\mathbf{x}} - A\boldsymbol{\mu}) \sim T_{q,n-1}^2$. This can be used for testing $H_0 : A\boldsymbol{\mu} = \mathbf{c}$.
 - If A is a matrix with rows which are contrasts so that $A\mathbf{1}_d = \mathbf{0}$, then

$$n\bar{\mathbf{x}}^T A^T (ASA^T)^{-1} A\bar{\mathbf{x}} = n\bar{\mathbf{x}}^T S^{-1} \bar{\mathbf{x}} - \frac{n(\bar{\mathbf{x}}^T S^{-1} \mathbf{1}_d)^2}{\mathbf{1}_d^T S^{-1} \mathbf{1}_d}.$$

12. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{n_1}$ be a random sample from $N_d(\boldsymbol{\mu}_1, \Sigma)$ and let $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{n_2}$ be an independent random sample from $N_d(\boldsymbol{\mu}_2, \Sigma)$, both distributions being nonsingular. Let $Q_1 = \sum_{i=1}^{n_1} (\mathbf{v}_i - \bar{\mathbf{v}})(\mathbf{v}_i - \bar{\mathbf{v}})^T$ and $Q_2 = \sum_{i=1}^{n_2} (\mathbf{w}_i - \bar{\mathbf{w}})(\mathbf{w}_i - \bar{\mathbf{w}})^T$. If $\boldsymbol{\theta} = \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$ and A is a $q \times d$ matrix of rank $q \leq d$, then:
 - $\bar{\mathbf{z}} = \bar{\mathbf{v}} - \bar{\mathbf{w}} \sim N_d(\boldsymbol{\theta}, (\frac{1}{n_1} + \frac{1}{n_2})\Sigma)$.
 - $Q = Q_1 + Q_2 \sim W_d(n_1 + n_2 - 2, \Sigma)$.
 - The statistic

$$\frac{n_1 n_2}{n_1 + n_2} (A\bar{\mathbf{z}} - A\boldsymbol{\theta})^T (AS_p A^T)^{-1} (A\bar{\mathbf{z}} - A\boldsymbol{\theta}) \sim T_{q, n_1 + n_2 - 2}^2,$$

where $S_p = Q/(n_1 + n_2 - 2)$. This can be used to test $H_0 : A\boldsymbol{\theta} = \mathbf{0}$. When A is a certain $(d-1) \times d$ contrast matrix, then the methodology relating to H_0 is called *profile analysis*.

Examples:

1. Suppose the rows of X are i.i.d. nonsingular $N_d(\mathbf{0}, \Sigma)$. Then $\bar{\mathbf{x}} = X^T \mathbf{1}_n / n$ and $W_1 = n\bar{\mathbf{x}}\bar{\mathbf{x}}^T = X^T A X$, where $A = \mathbf{1}_n \mathbf{1}_n^T / n$ is symmetric and idempotent of rank 1. Also, from Section 53.3, $W_2 = (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T = X^T C X$, where $C = I_n - A$ is symmetric and idempotent of rank $n-1$. Since $AC = \mathbf{0}$, we have from Fact 7 above that $W_1 \sim W_d(1, \Sigma)$, $W_2 \sim W_d(n-1, \Sigma)$, and W_1 and W_2 are statistically independent.

2. (cf. Fact 10) Suppose we have a group of 30 animals all subject to the same conditions, and the length of each animal is observed at d points in time $t_1, t_2, \dots, t_d$. The d lengths for the i th animal will give us a d -dimensional vector, $\mathbf{x}_i$ say, and we assume that the $\mathbf{x}_i$ are all i.i.d., as nonsingular $N_d(\boldsymbol{\mu}, \Sigma)$. We are interested in testing the hypothesis H_0 that the “growth curve” is a third degree polynomial so that $H_0 : \mu_j = \beta_0 + \beta_1 t_j + \beta_2 t_j^2 + \beta_3 t_j^3$ ($j = 1, 2, \dots, d$) or $H_0 : \boldsymbol{\mu} = K\boldsymbol{\beta}$, where the j th row of K is $[1, t_j, t_j^2, t_j^3]$ and $\boldsymbol{\beta} = [\beta_0, \beta_1, \beta_2, \beta_3]^T$ is a vector of unknown parameters. We then test H_0 by calculating $T_{\min}^2$ given in Fact 10, and this statistic, which has a Hotellings T^2 distribution when H is true, can then be converted to an F -statistic using Fact 8. If this F -value is significantly large we reject H_0 .
3. (cf. Fact 11) The lengths of the femur and humerus on the left- and right-hand sides are measured for $n = 100$ males aged over 40 years to see if men are symmetrical with respect to the lengths of these bones. For the i th male, we therefore have a four-dimensional observation vector $\mathbf{x}_i$. Assuming that the population of measurements is multivariate normal with mean $\boldsymbol{\mu} = [\mu_1, \mu_2, \mu_3, \mu_4]^T$, where μ_1 and μ_2 refer to the left side, then we are interested in testing the contrasts $\mu_1 - \mu_3 = 0$ and $\mu_2 - \mu_4 = 0$. Thus, A has rows $[1, 0, -1, 0]$ and $[0, 1, 0, -1]$, giving us $H_0 : A\mathbf{1}_4 = \mathbf{0}$. We can then test H_0 using $T_{\min}^2$ defined in Fact 11, and this T^2 statistic is then converted to an F -statistic, as in the previous example.

53.9 Multivariate Linear Model: Least Squares Estimation

Definitions:

Let Y be an $n \times d$ data matrix which comes from an experimental design giving rise to n observations, the rows of Y . Then we can use observations on the j th characteristic (variable) $\mathbf{y}^{(j)}$, the j th column of Y , to construct a linear regression model $\mathbf{y}^{(j)} = \boldsymbol{\theta}^{(j)} + \mathbf{u}^{(j)} = K\boldsymbol{\beta}^{(j)} + \mathbf{u}^{(j)}$ as in Section 52.2 (with K replaced by X there). Because the design is the same for each variable, the design matrix K will be independent of j ($j = 1, 2, \dots, d$), though the models will not be independent as the $\mathbf{y}^{(j)}$ are not independent. Putting all d regression models together, we get $Y = \Theta + U$, where $\Theta = KB$, B is $p \times d$ matrix of unknown parameters, K is $n \times p$ of rank r ($r \leq p$), and $U = [\mathbf{u}^{(1)}, \dots, \mathbf{u}^{(d)}] = [\mathbf{u}_1, \dots, \mathbf{u}_n]^T$. We shall assume that the $\mathbf{u}_i$ are a random sample from a distribution with mean $\mathbf{0}$ and covariance matrix $\Sigma = [\sigma_{ij}]$. Then $Y = KB + U$ is called a **multivariate linear model**. When $d = 1$, this reduces to the **univariate linear model** or **Gauss–Markov model**; see Chapter 52 and [Seb04, ch. 8].

Now, for the j th model, $\hat{\boldsymbol{\theta}}^{(j)} = P\mathbf{y}^{(j)}$ is the (ordinary) least squares estimate of $\boldsymbol{\theta}^{(j)}$, where $P = K(K^T K)^{-1}K^T$, and $\hat{\Theta} = PY$ is defined to be the **least squares estimate of Θ** . When $r = p$, then setting $\hat{\Theta} = K\hat{B}$ we have $\hat{B} = (K^T K)^{-1}K^T \hat{\Theta} = (K^T K)^{-1}K^T Y$, called the **least squares estimate of B** . If $r < p$, then $\hat{B}$ is not unique and is given by $\hat{B} = (K^T K)^{-}K^T Y$, where $(K^T K)^{-}$ is any generalized inverse of $K^T K$.

If K has less than full rank, then each of the d associated univariate models also has less than full rank. Moreover, $\mathbf{a}_i^T \boldsymbol{\beta}^{(j)}$ is estimable for each $i = 1, 2, \dots, q$ and each model $j = 1, 2, \dots, d$ if $\mathbf{a}_i \in \text{range}(K^T)$. Let $A = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_q]^T$. Combining these linear combinations we say that AB is **estimable** when $A = LK$ for some $q \times n$ matrix L .

Facts:

All the following facts can be found in [Seb04, Ch. 8].

1. If $K^T = [\mathbf{k}_1, \dots, \mathbf{k}_n]$ then, transposing the model, $\mathbf{y}_i = B^T \mathbf{k}_i + \mathbf{u}_i$.
2. The cross-covariance $\text{cov}(\mathbf{y}_r, \mathbf{y}_s) = \text{cov}(\mathbf{u}_r, \mathbf{u}_s) = \delta_{rs} \Sigma$, where $\delta_{rs} = 1$ when $r = s$ and 0, otherwise.
3. The cross-covariance $\text{cov}(\mathbf{y}^{(j)}, \mathbf{y}^{(k)}) = \text{cov}(\mathbf{u}^{(j)}, \mathbf{u}^{(k)}) = \sigma_{jk} I_d$ for all $j, k = 1, \dots, d$.
4. If K has full rank p , then $\hat{\boldsymbol{\beta}}^{(j)} = (K^T K)^{-1}K^T \mathbf{y}^{(j)}$ and $\text{cov}(\hat{\boldsymbol{\beta}}^{(j)}, \hat{\boldsymbol{\beta}}^{(k)}) = \sigma_{jk}(K^T K)^{-1}$ (all $j, k = 1, \dots, d$).

5. Let $F(\Theta) = (Y - \Theta)^T(Y - \Theta)$. Then:
 - $F(\hat{\Theta}) = Y^T(I_n - P)Y = U^T(I_n - P)U$.
 - $E[U^T(I_n - P)U] = (n - r)\Sigma$.
6. $F(\Theta) - F(\hat{\Theta})$ is positive semidefinite for all $\Theta = XB$, and equal to 0 if and only if $\Theta = \hat{\Theta}$. We say that $\hat{\Theta}$ is the minimum of $F(\Theta)$. Then:
 - $\text{tr}F(\Theta) \geq \text{tr}F(\hat{\Theta})$.
 - $\det F(\Theta) \geq \det F(\hat{\Theta})$.
 - $\|F(\Theta)\| \geq \|F(\hat{\Theta})\|$, where $\|A\| = \{\text{tr}(AA^T)\}^{1/2}$.

Any of these three results could be used as a definition of $\hat{\Theta}$.
7. *Multivariate Gauss–Markov Theorem.* If $\phi = \sum_{j=1}^d \mathbf{h}_j^T \boldsymbol{\theta}^{(j)}$, a linear combination of all the elements of Θ , then $\hat{\phi} = \sum_{j=1}^d \mathbf{h}_j^T \hat{\boldsymbol{\theta}}^{(j)}$ is the linear unbiased estimate with minimum variance or BLUE (best linear unbiased estimate) of ϕ . (See also Chapter 52.)

Examples:

1. [Seb04, Sec. 8.6.4] By setting $V^T = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{n_1}]$, $W^T = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{n_2}]$, and $Y = \begin{bmatrix} V \\ W \end{bmatrix}$, we see that the *two-sample problem* mentioned in Fact 12 of Section 53.8 is a special case of the multivariate model with

$$XB = \begin{bmatrix} \mathbf{1}_{n_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1}_{n_2} \end{bmatrix} \begin{bmatrix} \boldsymbol{\mu}_1^T \\ \boldsymbol{\mu}_2^T \end{bmatrix}.$$

53.10 Multivariate Linear Model: Statistical Inference

Definitions:

Let $Y = \Theta + U$, where $\Theta = KB$, be a multivariate linear model. We now assume that the underlying distribution (of the $\mathbf{u}_i$) is a nonsingular multivariate normal distribution $N_d(\mathbf{0}, \Sigma)$.

Facts:

The following facts appear in [Seb04, Sec. 8.6].

1. The likelihood function for Y , i.e., the pdf of $\text{vec } Y$, can be expressed in the form

$$(2\pi)^{-nd/2} (\det \Sigma)^{-n/2} \exp\left\{\text{tr}\left[-\frac{1}{2}(Y - \Theta)^T \Sigma^{-1}(Y - \Theta)\right]\right\}.$$

2. The maximum likelihood estimates of Σ and Θ that maximize the likelihood function are $\hat{\Theta}$ (the least squares estimate) and $\hat{\Sigma} = E/n$, where $E = (Y - \hat{\Theta})^T(Y - \hat{\Theta})$; E is usually called the residual matrix or error matrix. If the $n \times p$ matrix K has full rank (i.e., rank p), then the maximum likelihood estimate of B is $\hat{B}$.
3. Let E be the residual matrix and assume $n - r \geq d$. Then
 - E is positive definite with probability 1.
 - $E \sim W_d(n - r, \Sigma)$.
 - E is statistically independent of $\hat{\Theta}$, and of $\hat{B}$ if K has full rank.
 - The maximum value of the likelihood function is $(2\pi)^{-nd/2} (\det \hat{\Sigma})^{-n/2} e^{-nd/2}$.
 - If K has full rank, then $\hat{\boldsymbol{\beta}}^{(j)} \sim N_n(\boldsymbol{\beta}^{(j)}, \sigma_{jj}(K^T K)^{-1})$.

4. Suppose that K has rank $r < p$. Let A be a known $q \times p$ matrix of rank q and let AB be estimable. We are interested in testing $H_0 : AB = C$, where C is known.

- The minimum, E_H say, of $(Y - KB)^T(Y - KB)$ subject to $AB = C$ occurs when B equals

$$\widehat{B}_H = \widehat{B} - (K^T K)^{-1} A^T [A(K^T K)^{-1} A^T]^{-1} (A\widehat{B} - C).$$

Although $\widehat{B}_H$ is not unique, $\widehat{\Theta}_H = K\widehat{B}_H$ is unique. Moreover, $E_H = (Y - \widehat{\Theta}_H)^T(Y - \widehat{\Theta}_H)$ is positive definite with probability 1.

- $H = E_H - E = (A\widehat{B} - C)^T [A(K^T K)^{-1} A^T]^{-1} (A\widehat{B} - C)$ is positive definite with probability 1, and H and E are statistically independent (both irrespective of whether H_0 is true or not).
 - $E(H) = q\Sigma + (AB - C)^T [A(X^T X)^{-1} A^T]^{-1} (AB - C) = q\Sigma + D$, where D is positive definite; D is zero when H_0 is true. This means that $E(H)$, and, therefore, H itself, tends to be “inflated” when H_0 is false so that a test statistic for H_0 can be based on a suitable function of H .
 - When H_0 is true, $H \sim W_d(q, \Sigma)$.
 - Let $E_H^{1/2}$ be the positive definite square root of E_H . Then when H_0 is true, $V = E_H^{-1/2} H E_H^{-1/2}$, a scaled function of H , has a d -dimensional multivariate Beta distribution with parameters q and $n - r$.
5. [Seb04, Sec. 8.6.2] Four different criteria are usually computed for testing H_0 , and these are essentially based on the eigenvalues of V given above, which measure the scaled magnitude of H in some sense. They detect different kinds of departures from H_0 and are all significantly large when H_0 is false.
- Roy’s maximum root statistic $\phi_{\max}$, the maximum eigenvalue of HE^{-1} .
 - Wilks’ Lambda or likelihood ratio statistic $\Lambda = (\det E / \det E_H)^{n/2}$.
 - Lawley–Hotelling trace statistic $(n - r)\text{tr}(HE^{-1})$.
 - Pillai’s trace statistic $\text{tr}(HE_H^{-1})$.

Example:

1. [Seb04, Sec. 8.7.2] To test the general linear hypothesis $H_0 : ABD = \mathbf{0}$, where A is $q \times p$ of rank $q \leq p$ and D is $d \times v$ of rank $v \leq d$, we let $Y_D = YD$ so that the linear model $Y = KB + U$ is transformed to

$$Y_D = KBD + UD = K\Lambda + U_0,$$

say, where the rows of U_0 are i.i.d. $N_v(\mathbf{0}, D^T \Sigma D)$. Then H_0 becomes $A\Lambda = \mathbf{0}$ and

- H becomes $H_D = D^T H D = (A\widehat{B}D)^T [A(K^T K)^{-1} A^T]^{-1} A\widehat{B}D$ and E becomes $E_D = D^T E D \sim W_v(n - r, D^T \Sigma D)$.
- When $ABD = \mathbf{0}$ is true, then $H_D \sim W_v(q, D^T \Sigma D)$.

We can now use Facts 4 and 5 above to test H_0 using H_D and E_D instead of H and E . This hypothesis arises in carrying out a so-called profile analysis of more than two populations.

53.11 Metric Multidimensional Scaling

Definitions:

Given a set of n objects, a **proximity measure** d_{rs} is a measure of the “closeness” of objects r and s ; here, “closeness” does not necessarily refer to physical distance. We shall consider only one such measure as there are several. A proximity d_{rs} is called a **(symmetric) dissimilarity** if $d_{rr} = 0$, $d_{rs} \geq 0$, and $d_{rs} = d_{sr}$,

for all $r, s = 1, 2, \dots, n$; the matrix $D = [d_{rs}]$ is called a **dissimilarity matrix**. We say that D is **Euclidean** if there exists a p -dimensional configuration of points $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n$ for some p such that all interpoint Euclidean distances satisfy $\|\mathbf{y}_r - \mathbf{y}_s\| = d_{rs}$.

Facts:

1. Let $A = [a_{ij}]$ be a symmetric $n \times n$ matrix, where $a_{rs} = -\frac{1}{2}d_{rs}^2$. Define $b_{rs} = a_{rs} - \bar{a}_{r.} - \bar{a}_{.s} + \bar{a}_{..}$, where $\bar{a}_{r.}$ is the average of the elements of A in the r th row, $\bar{a}_{.s}$ is the average for the s th column, and $\bar{a}_{..}$ is the average of all the elements. Hence, $B = [b_{rs}] = CAC$, where $C = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T$ is the usual centering matrix. Then $D = [d_{rs}]$ is Euclidean if and only if B is positive semidefinite; see [Seb04, p. 236].
2. If D is not Euclidean, then some eigenvalues of B will be negative. However, if the first k eigenvalues are comparatively large and positive, and the remaining positive or negative eigenvalues are near zero, then the rows of $Y_k = [\mathbf{y}^{(1)}, \mathbf{y}^{(2)}, \dots, \mathbf{y}^{(k)}]$ will give a reasonable configuration. If the original objects are d -dimensional points $\mathbf{x}_i$ ($i = 1, 2, \dots, n$) to begin with so that $\|\mathbf{x}_r - \mathbf{x}_s\|^2 = d_{rs}^2$, then D is Euclidean and the n rows of Y_k will give a k -dimensional reduction of a d -dimensional system of points.

Example:

1. The migration pattern that occurred with the colonization of islands in the Pacific Ocean can be investigated linguistically and values of d_{rs} can be constructed based on linguistic information. For example, the proportion p_{rs} of the words for, say, 50 items that islands r and s have in common can be used as a measure of similarity (with $p_{rr} = 1$), which we can convert into a dissimilarity $d_{rs} = (2 - 2p_{rs})^{1/2}$. Then $a_{rs} = -\frac{1}{2}d_{rs}^2 = p_{rs} - 1$ and, when computing b_{rs} , the -1 drops out so that we can leave it out and set $a_{rs} = p_{rs}$. If A is positive semidefinite then so is B , which implies that D is Euclidean. Using B , we can then find a Y_k from which we can construct a lower dimensional map to see which islands are closest together linguistically. The same method can also be applied to blood groups using a different d_{rs} .

Acknowledgment

We are grateful to Ka Lok Chu, Jarkko Isotalo, Evelyn Mathason Styan, Kimmo Vahkalahti Andrei Volodin, and Douglas P. Wiens for their help. The research was supported in part by the National Sciences and Engineering Council of Canada.

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54

Markov Chains

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Markov chains are encountered in several applications arising in different contexts, and model many real problems which evolve in time. Throughout, we denote by $P[X = j]$ the probability that the random variable X takes the value j , and by $P[X = j|Y = i]$ the conditional probability that X takes the value j , given that the random variable Y takes the value i . Moreover, we denote by $E[X]$ the expected value of the random variable X and by $E[X|A]$ the conditional expectation of X , given the event A .

54.1 Basic Concepts

Definitions:

Given a denumerable set E , a **discrete stochastic process** on E is a family $\{X_t : t \in T\}$ of random variables X_t indexed by some denumerable set T and with values in E , i.e., $X_t \in E$ for all $t \in T$. Here, E is the **state space**, and T is the **time space**.

The discrete stochastic process $\{X_n : n \in \mathbb{N}\}$ is a **Markov chain** if

$$P[X_{n+1} = j_{n+1} | X_0 = j_0, X_1 = j_1, \dots, X_n = j_n] = P[X_{n+1} = j_{n+1} | X_n = j_n], \quad (54.1)$$

for any $(n+2)$ -tuple of states $\{j_0, \dots, j_{n+1}\} \in E$, and for all time $n \in \mathbb{N}$.

A Markov chain $\{X_n : n \in \mathbb{N}\}$ is **homogeneous** if

$$P[X_{n+1} = j | X_n = i] = P[X_1 = j | X_0 = i], \quad (54.2)$$

for all states $i, j \in E$ and for all time $n \in \mathbb{N}$.

Given a homogeneous Markov chain $\{X_n : n \in \mathbb{N}\}$ we define the **transition matrix** of the Markov chain to be the matrix $P = [p_{i,j}]_{i,j \in E}$ such that

$$p_{i,j} = P[X_1 = j | X_0 = i], \quad \text{for all } i, j \text{ in } E.$$

Throughout, unless differently specified, for the sake of notational simplicity we will indicate with the term **Markov chain** a homogeneous Markov chain.

A **finite Markov chain** is a Markov chain with finite space state E ; an **infinite Markov chain** is a Markov chain with infinite space state E .

A row vector $\pi = (\pi_i)_{i \in E}$ such that

$$\pi P = \pi \quad (54.3)$$

is an **invariant vector**.

If the invariant vector π is such that

$$\pi_i \geq 0 \text{ for all } i, \quad \text{and} \quad \sum_{i \in E} \pi_i = 1, \quad (54.4)$$

then π is an **invariant probability vector**, or **stationary distribution**. The definition of stationary distribution extends the definition given in Section 9.4 to the case of an infinite matrix P .

Facts:

1. The Markov property in Equation (54.1) means that the state X_n of the system at time n is sufficient to determine which state might be occupied at time $n + 1$, and the past history $X_0, X_1, \dots, X_{n-1}$ does not influence the future state at time $n + 1$.
2. In a homogeneous Markov chain, the property in Equation (54.2) means that the laws which govern the evolution of the system are independent of the time n ; therefore, the evolution of the Markov chain is ruled by the transition matrix P , whose (i, j) th entry represents the probability to change from state i to state j in one time unit.
3. The number of rows and columns of the transition matrix P is equal to the cardinality of E . In particular, if the set E is finite, P is a finite matrix (see Examples 1 and 5); if the set E is infinite, P is an infinite matrix, i.e., a matrix with an infinite number of rows and columns (see Examples 2–4).
4. The matrix P is row stochastic, i.e., it is a matrix with nonnegative entries such that $\sum_{j \in E} P_{i,j} = 1$ for any $i \in E$, i.e., the sum of the entries on each row is equal to 1.
5. If $|E| < \infty$, the matrix P has spectral radius 1. Moreover, the vector $\mathbf{1}_{|E|}$ is a right eigenvector of P corresponding to the eigenvalue 1; any nonzero invariant vector is a left eigenvector of P corresponding to the eigenvalue 1; the invariant probability vector π is a nonnegative left eigenvector of P corresponding to the eigenvalue 1, normalized so that $\pi \mathbf{1}_{|E|} = 1$.
6. In the analysis of Markov chains, we may encounter matrix products $A = BC$, where $B = [b_{i,j}]_{i,j \in E}$ and $C = [c_{i,j}]_{i,j \in E}$ are nonnegative matrices such that $\sum_{j \in E} b_{i,j} \leq 1$ and $\sum_{j \in E} c_{i,j} \leq 1$ for any $i \in E$ (see, for instance, Fact 7 below). The (i, j) th entry of A , given by $a_{i,j} = \sum_{h \in E} b_{i,h} c_{h,j}$, is well defined also if the set E is infinite, since $0 \leq \sum_{h \in E} b_{i,h} c_{h,j} \leq \sum_{h \in E} b_{i,h} \leq 1$. Moreover, A is a nonnegative matrix such that $\sum_{j \in E} a_{i,j} \leq 1$ for any $i \in E$. Indeed, $\sum_{j \in E} a_{i,j} = \sum_{j \in E} \sum_{h \in E} b_{i,h} c_{h,j} = \sum_{h \in E} b_{i,h} \sum_{j \in E} c_{h,j} \leq \sum_{h \in E} b_{i,h} \leq 1$. Similarly, the product $\mathbf{v} = \mathbf{u}B$, where $\mathbf{u} = (u_i)_{i \in E}$ is a nonnegative row vector such that $\sum_{i \in E} u_i \leq 1$, is well defined also in the case where E is infinite; moreover, $\mathbf{v} = (v_i)_{i \in E}$ is a nonnegative vector such that $\sum_{i \in E} v_i \leq 1$.
7. [Nor99, Theorem 1.1.3] The dynamic behavior of a homogeneous Markov chain is completely characterized by the transition matrix P :

$$P[X_{n+k} = j | X_n = i] = (P^k)_{i,j},$$

for all times $n \geq 0$, all intervals of time $k \geq 0$, and all pairs of states i and j in E .

8. In addition to the system dynamics, one must choose the starting point X_0 . Let $\pi^{(0)} = (\pi_i^{(0)})_{i \in E}$ be a probability distribution on E , i.e., a nonnegative row vector such that the sum of its components is equal to one. Assume that $\pi_i^{(0)} = P[X_0 = i]$, and define the row vector $\pi^{(n)} = (\pi_i^{(n)})_{i \in E}$ to be the probability vector of the Markov chain at time $n \geq 1$, that is, $\pi_i^{(n)} = P[X_n = i | X_0]$.

Then

$$\pi^{(n+1)} = \pi^{(n)} P, \quad n \geq 0, \quad (54.5)$$

$$\pi^{(n)} = \pi^{(0)} P^n, \quad n \geq 0. \quad (54.6)$$

9. If the initial distribution $\pi^{(0)}$ coincides with the invariant probability vector π , then $\pi^{(n)} = \pi$ for any $n \geq 0$.
10. In certain applications (see for instance, Example 5) we are interested in the asymptotic behavior of the Markov chain. In particular, we would like to compute, if it exists, the vector $\lim_{n \rightarrow \infty} \pi^{(n)}$. From Equation (54.5) of Fact 8 one deduces that, if such limit exists, it coincides with the invariant probability vector, i.e., $\lim_{n \rightarrow \infty} \pi^{(n)} = \pi$.

Examples:

1. *Random walk on $\{0, 1, \dots, k\}$:* Consider a particle which moves on the interval $[0, k]$ in unit steps at integer instants of time; let $X_n \in \{0, 1, \dots, k\}$, $n \geq 0$, be the position of the particle at time n and let $0 < p < 1$. Assume that, if the particle is in the open interval $(0, k)$, at the next unit time it will move to the right with probability p and to the left with probability $q = 1 - p$; if the particle is in position 0, it will move to the right with probability 1; if the particle is in position k , it will move to the left with probability 1. Clearly, the discrete stochastic process $\{X_n : n \in \mathbb{N}\}$ is a homogeneous Markov chain with space state the set $E = \{0, 1, \dots, k\}$. The transition matrix is the $(k + 1) \times (k + 1)$ matrix $P = [p_{i,j}]_{i,j=0,\dots,k}$, given by

$$P = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ q & 0 & p & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & q & 0 & p \\ 0 & \dots & 0 & 1 & 0 \end{bmatrix}.$$

From Equation (54.6) of Fact 8 one has that, if the particle is in position 0 at time 0, the probability vector at time $n = 10$ is the vector

$$[\pi_0^{(10)}, \pi_1^{(10)}, \dots, \pi_k^{(10)}] = [1, 0, \dots, 0] P^{10}.$$

2. *Random walk on $\mathbb{N}$:* If we allow the particle to move on $\mathbb{N}$, we have a homogeneous Markov chain with state space the set $E = \mathbb{N}$. The transition matrix is semi-infinite and is given by

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 & \dots \\ q & 0 & p & 0 & \ddots \\ 0 & q & 0 & p & \ddots \\ 0 & 0 & q & 0 & \ddots \\ \vdots & \ddots & \ddots & \ddots & \ddots \end{bmatrix}.$$

3. *Random walk on $\mathbb{Z}$* : If we allow the particle to move on $\mathbb{Z}$, we still have a homogeneous Markov chain, with state space the set $E = \mathbb{Z}$. The transition matrix is bi-infinite and is given by

$$P = \begin{bmatrix} \ddots & \ddots & \ddots & \ddots & \ddots \\ \ddots & 0 & p & 0 & \ddots \\ \ddots & q & 0 & p & \ddots \\ \ddots & 0 & q & 0 & \ddots \\ \ddots & \ddots & \ddots & \ddots & \ddots \end{bmatrix}.$$

4. *A simple queueing system* [BLM05, Example 1.3]: Simple queues consist of one server which attends to one customer at a time, in order of their arrivals. We assume that time is discretized into intervals of length one, that a random number of customers join the system during each interval, that customers do not leave the queue, and that the server removes one customer from the queue at the end of each interval, if there is any. Defining α_n as the number of new arrivals during the interval $[n-1, n)$ and X_n as the number of customers in the system at time n , we have

$$X_{n+1} = \begin{cases} X_n + \alpha_{n+1} - 1 & \text{if } X_n + \alpha_{n+1} \geq 1 \\ 0 & \text{if } X_n + \alpha_{n+1} = 0. \end{cases}$$

If $\{\alpha_n\}$ is a collection of independent random variables, then X_{n+1} is conditionally independent of $X_0, \dots, X_{n-1}$ if X_n is known. If, in addition, the α_n 's are identically distributed, then $\{X_n\}$ is homogeneous. The state space is $\mathbb{N}$ and the transition matrix is

$$P = \begin{bmatrix} q_0 + q_1 & q_2 & q_3 & q_4 & \cdots \\ q_0 & q_1 & q_2 & q_3 & \ddots \\ & q_0 & q_1 & q_2 & \ddots \\ 0 & & \ddots & \ddots & \ddots \end{bmatrix},$$

where q_i is the probability $P[\alpha = i]$ that i new customers join the queue during a unit time interval, α denoting any of the identically distributed random variables α_n . Markov chains having transition matrix of the form

$$P = \begin{bmatrix} B_1 & B_2 & B_3 & B_4 & \cdots \\ A_0 & A_1 & A_2 & A_3 & \ddots \\ & A_0 & A_1 & A_2 & \ddots \\ 0 & & \ddots & \ddots & \ddots \end{bmatrix},$$

where $A_i, B_{i+1}, i \geq 0$, are nonnegative $k \times k$ matrices, are called M/G/1-type Markov chains, and model a large variety of queueing problems. (See [Neu89], [LR99], and [BLM05].)

5. *Search engines* (see Section 63.5, [PBM99], [ANT02] and [Mol04, Section 2.11]): PageRank is used by Google to sort, in order of relevance, the pages on the Web that match the query of the user. From the Web page *Google Technology* at <http://www.google.com/technology/>: “The heart of our software is PageRankTM, a system for ranking Web pages developed by our founders Larry Page and Sergey Brin at Stanford University. And while we have dozens of engineers working to improve every aspect of Google on a daily basis, PageRank continues to provide the basis for all

of our Web search tools.” Surfing on the Web is seen as a random walk, where either one starts from a Web page and goes from one page to the next page by randomly following a link (if any), or one simply chooses a random page from the Web. Let E be the set of Web pages that can be reached by following a sequence of hyperlinks starting from a given Web page. If k is the number of Web pages, i.e., $k = |E|$, the connectivity matrix is defined as the $k \times k$ matrix $G = [g_{i,j}]$ such that $g_{i,j} = 1$ if there is a hyperlink from page i to page j , and zero otherwise. Let $r_i = \sum_j g_{i,j}$ and $c_i = \sum_j g_{j,i}$ be the row and column sums of G ; the quantities r_i and c_i are called the out-degree and the in-degree, respectively, of the page i . Let $0 < q < 1$ and let $P = [p_{i,j}]$ be the $k \times k$ stochastic matrix such that $p_{i,j} = qg_{i,j}/r_i + (1-q)/k$, $i, j = 1, \dots, k$. The value q is the probability that the random walk on E follows a link and, therefore, $1-q$ is the probability that an arbitrary page is chosen. The matrix P is the transition matrix of the Markov chain that models the random walk on E . The importance of a Web page is related to the probability to reach such page during this random walk as the time tends to infinity. Therefore, Google’s PageRank is determined by the invariant probability vector π of P : The larger the value of an entry π_i of π , the higher the relevance of the i th Web page in the set E .

54.2 Irreducible Classes

Some of the Definitions and Facts given in this section extend the corresponding Definitions and Facts of Chapter 9 and Chapter 29. Indeed, in these chapters, it is assumed that the matrices have a finite size, while in the framework of Markov chains we may encounter infinite matrices.

Definitions:

The **transition graph** of a Markov chain with transition matrix P is the digraph of P , $\Gamma(P)$. That is, the digraph defined as follows: To each state in E there corresponds a vertex of the digraph and one defines a directed arc from vertex i to vertex j for each pair of states such that $p_{i,j} > 0$. More information about digraphs can be found in Chapter 9 and Chapter 29.

A **closed walk** in a digraph is a walk in which the first vertex equals the last vertex.

State i **leads to** j (or i **has access to** j) if there is a walk from i to j in the transition graph. States i and j **communicate** if i leads to j and j leads to i .

A Markov chain is called **irreducible** if the transition graph is strongly connected, i.e., if all the states communicate. A Markov chain is called **reducible** if it is not irreducible.

The strongly connected components of the transition graph are the **communicating classes of states**, or of the Markov chain. Communicating classes are also called **access equivalence classes** or **irreducible classes**.

A communicating class C is a **final class** if for every state i in C , there is no state j outside of C such that i leads to j . If, on the contrary, there is a state in C that leads to some state outside of C , the class is a **passage class**. A single state that forms a final class by itself is **absorbing**. In Section 9.4, passage classes are called transient classes, and final classes are called ergodic classes; in fact, transient and ergodic states of Markov chains will be introduced in Section 54.3, and for finite Markov chains the states in a passage class are transient, and the states in a final class are ergodic, cf. Section 54.4.

A state i is **periodic** with period $\delta \geq 2$ if all closed walks through i in the transition graph have a length that is a multiple of δ . A state i is **aperiodic** if it is not periodic.

A Markov chain is **periodic** with period δ if all states are periodic and have the same period δ .

Facts:

1. A Markov chain is irreducible if and only if the transition matrix P is irreducible, i.e., if P is not permutation similar to a block triangular matrix, cf. Section 9.2, Section 27.1.
2. If we adopt the convention that each state communicates with itself, then the relation *communicates* is an equivalence relation and the communicating classes are the equivalence classes of this relation, cf. Section 9.1.

3. A Markov chain is irreducible if and only if the states form one single communicating class, cf. Section 9.1, Section 9.2.
4. If a Markov chain with transition matrix P has $K \geq 2$ communicating classes, denoted by $C_1, C_2, \dots, C_K$, then the states may be permuted so that the transition matrix $P' = \Pi P \Pi^T$ associated with the permuted states is block triangular:

$$P' = \begin{bmatrix} P_{1,1} & P_{1,2} & \dots & P_{1,K} \\ & P_{2,2} & \ddots & \vdots \\ & & \ddots & P_{K-1,K} \\ 0 & & & P_{K,K} \end{bmatrix}$$

where $P_{i,j}$ is the submatrix of transition probabilities from the states of C_i to C_j , the diagonal blocks are irreducible square matrices, and Π is the permutation matrix associated with the rearrangement, cf. Section 9.2.

5. [Çin75, Theorem (3.16), Chap. 5] Periodicity is a class property and all states in a communicating class have the same period. Thus, for irreducible Markov chains, either all states are aperiodic, or all have the same period δ , which we may call the period of the Markov chain itself.

Examples:

1. Figure 54.1 is the transition graph associated with the Markov chain on $E = \{1, 2, 3\}$ with transition matrix

$$P = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}.$$

The two sets $C_1 = \{1\}$ and $C_2 = \{2, 3\}$ are two communicating classes. C_2 is a passage class, while C_1 is a final class, therefore state 1 is absorbing.

2. Figure 54.2 is the transition graph associated with the Markov chain on $E = \{k \in \mathbb{N} : k \geq 1\}$ with transition matrix having following structure:

$$P = \begin{bmatrix} 0 & * & & & & & 0 \\ 0 & 0 & * & & & & \\ * & 0 & 0 & * & & & \\ 0 & 0 & 0 & 0 & * & & \\ 0 & 0 & * & 0 & 0 & * & \\ 0 & 0 & 0 & 0 & 0 & 0 & \ddots \\ 0 & 0 & 0 & 0 & * & 0 & \ddots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots \end{bmatrix},$$

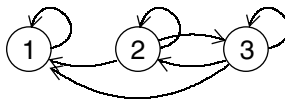


FIGURE 54.1 Transition graph of the Markov chain of Example 1.

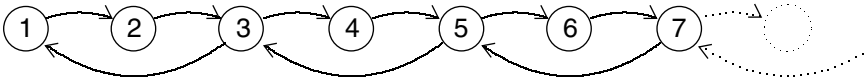


FIGURE 54.2 Graph of an infinite irreducible periodic Markov chain of period 3.

where “*” denotes a nonzero element. This is an example of a periodic irreducible Markov chain of period 3.

54.3 Classification of the States

Definitions:

Let T_j be the time of the first visit to j , without taking into account the state at time 0, i.e.,

$$T_j = \min\{n \geq 1 : X_n = j\}.$$

Define

$$f_j = P[T_j < \infty | X_0 = j],$$

that is, f_j is the probability that, starting from j , the Markov chain returns to j in a finite time.

A state $j \in E$ is **transient** if $f_j < 1$.

A state $j \in E$ is **recurrent** if $f_j = 1$. A recurrent state $j \in E$ is **positive recurrent** if the expected return time $E[T_j | X_0 = j]$ is finite; it is **null recurrent** if the expected return time $E[T_j | X_0 = j]$ is infinite. Positive recurrent states are also called **ergodic**.

A Markov chain is **positive/null recurrent** or **transient** if all its states are positive/null recurrent or transient, respectively.

A **regular Markov chain** is a positive recurrent Markov chain that is aperiodic.

The matrix $R = \sum_{n=0}^{\infty} P^n$ is the **potential matrix** of the Markov chain.

Facts:

1. Transient states may be visited only a finite number of times by the Markov chain. On the contrary, once the Markov chain has visited one recurrent state, it will return to it over and over again; if j is null recurrent the expected time between two successive visits to j is infinite.
2. [Çin75, Cor. (2.13), Chap. 5] For the potential matrix $R = [r_{i,j}]$ one has $r_{j,j} = 1/(1 - f_j)$, where we set $1/0 = \infty$.
3. [Çin75, Sec. 3, Chap. 5] The (j, j) th entry of R is the expected number of returns of the Markov chain to state j . A state j is recurrent if and only if $r_{j,j} = \infty$. A state j is transient if and only if $r_{j,j} < \infty$.
4. [Nor99, Theorems 1.5.4, 1.5.5, 1.7.7] The nature of a state is a class property. More specifically, the states in a passage class are transient; in a final class the states are either all positive recurrent, or all null recurrent, or all transient.
5. [Nor99, Theorem 1.5.6] If a final class contains a finite number of states only, then all its states are positive recurrent.
6. From Facts 4 and 5 one has that, for a communicating class with a finite number of states, either all the states are transient or all the states are positive recurrent.
7. From Fact 4 above, if a Markov chain is irreducible, the states are either all positive recurrent, or all null recurrent, or all transient. Therefore, an irreducible Markov chain is either positive recurrent or null recurrent or transient.

8. [Nor99, Secs. 1.7 and 1.8] Assume that the Markov chain is irreducible. The Markov chain is positive recurrent if and only if there exists a strictly positive invariant probability vector, that is, a row vector $\pi = (\pi_i)_{i \in E}$ such that $\pi_i > 0$ that satisfies Equations (54.3), (54.4) of Section 54.1. The invariant vector π is unique among nonnegative vectors, up to a multiplicative constant.
9. [Nor99, Secs. 1.7 and 1.8], [BLM05, Sec. 1.5] If the Markov chain is irreducible and null recurrent, there exists a strictly positive invariant vector, unique up to a multiplicative constant, such that the sum of its elements is not finite. Thus, there always exists an invariant vector for the transition matrix of a recurrent Markov chain. Some transient Markov chains also have an invariant vector (with infinite sum of the entries, like in the null recurrent case) but some do not.
10. [Çin75, Theorem (3.2), Chap. 5], [Nor99, Secs. 1.7 and 1.8] If j is a transient or null recurrent state, then for any $i \in E$, $\lim_{n \rightarrow \infty} (P^n)_{i,j} = 0$. If j is a positive recurrent and aperiodic state, then $\lim_{n \rightarrow \infty} (P^n)_{j,j} > 0$. If j is periodic with period δ , then $\lim_{n \rightarrow \infty} (P^{n\delta})_{j,j} > 0$.
11. [Nor99, Secs. 1.7 and 1.8] Assume that the Markov chain is irreducible, aperiodic, and positive recurrent. Then $\lim_{n \rightarrow \infty} (P^n)_{i,j} = \pi_j > 0$ for all j , independently of i , where $\pi = (\pi_i)_{i \in E}$ is the stationary distribution.
12. Facts 3, 4, 6, and 10 provide a criterium to classify the states. First identify the communicating classes. If a communicating class contains a finite number of states, the states are all positive recurrent if the communicating class is final; the states are all transient if the communicating class is a passage class. If a communicating class has infinitely many states, we apply Fact 3 to determine if they are recurrent or transient; for recurrent states we use Fact 10 to determine if they are null or positive recurrent.

Examples:

1. Let P be the transition matrix of Example 1 of Section 54.2. Observe that with positive probability the Markov chain moves from state 2 or 3 to state 1, and when the Markov chain will be in state 1, it will remain there forever. Indeed, according to Facts 4 and 5, states 2 and 3 are transient since they belong to a passage class, and state 1 is positive recurrent since it is absorbing. Moreover, if we partition the matrix P into the 2×2 block matrix

$$P = \begin{bmatrix} 1 & 0 \\ A & B \end{bmatrix},$$

where

$$A = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{4} \end{bmatrix}, \quad B = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{4} \end{bmatrix},$$

we may easily observe that

$$P^n = \begin{bmatrix} 1 & 0 \\ \sum_{i=0}^{n-1} B^i A & B^n \end{bmatrix}.$$

Since $\|B\|_1 = \frac{5}{6}$, then $\rho(B) < 1$, and therefore $\lim_{n \rightarrow \infty} B^n = 0$ and $\sum_{n=0}^{\infty} B^n = (I - B)^{-1}$. A simple computation leads to

$$\lim_{n \rightarrow \infty} P^n = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad R = \sum_{n=0}^{\infty} P^n = \begin{bmatrix} \infty & 0 & 0 \\ \infty & \frac{9}{4} & 1 \\ \infty & \frac{3}{2} & 2 \end{bmatrix},$$

in accordance with Facts 3 and 10.

2. [BLM05, Ex. 1.19] The transition matrix

$$P = \begin{bmatrix} 0 & 1 & & & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & & \\ & \frac{1}{2} & 0 & \frac{1}{2} & \\ 0 & & \ddots & \ddots & \ddots \end{bmatrix}$$

is irreducible, and $\pi = [\frac{1}{2}, 1, 1, \dots]$ is an invariant vector. The vector π has “infinite” mass, that is, the sum of its components is infinite. In fact, the Markov chain is actually null recurrent (see [BLM05, Sec. 1.5]).

3. [BLM05, Ex. 1.20] For the transition matrix

$$P = \begin{bmatrix} 0 & 1 & & & 0 \\ \frac{1}{4} & 0 & \frac{3}{4} & & \\ & \frac{1}{4} & 0 & \frac{3}{4} & \\ 0 & & \ddots & \ddots & \ddots \end{bmatrix}$$

one has $\pi P = \pi$ with $\pi = [1, 4, 12, 36, 108, \dots]$. The vector π has unbounded elements. In this case the Markov chain is transient.

4. [BLM05, Ex. 1.21] For the transition matrix

$$P = \begin{bmatrix} 0 & 1 & & & 0 \\ \frac{3}{4} & 0 & \frac{1}{4} & & \\ & \frac{3}{4} & 0 & \frac{1}{4} & \\ 0 & & \ddots & \ddots & \ddots \end{bmatrix}$$

one has $\pi P = \pi$ with $\pi = \frac{4}{3}[\frac{1}{4}, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \dots]$ and $\sum_i \pi_i = 1$. In this case the Markov chain is positive recurrent by Fact 8.

54.4 Finite Markov Chains

The transition matrix of a finite Markov chain is a finite dimensional stochastic matrix; the reader is advised to consult Section 9.4 for properties of finite stochastic matrices.

Definitions:

A $k \times k$ matrix A is **weakly cyclic of index** δ if there exists a permutation matrix Π such that $A' = \Pi A \Pi^T$ has the block form

$$A' = \begin{bmatrix} 0 & 0 & \dots & 0 & A_{1,\delta} \\ A_{2,1} & 0 & \ddots & & 0 \\ 0 & A_{3,2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 & 0 \\ 0 & \dots & 0 & A_{\delta,\delta-1} & 0 \end{bmatrix},$$

where the zero diagonal blocks are square.

Facts:

1. If P is the transition matrix of a finite Markov chain, there exists an invariant probability vector π . If P is irreducible, the vector π is strictly positive and unique, cf. Section 9.4.
2. For a finite Markov chain no state is null recurrent, and not all states are transient. The states belonging to a final class are positive recurrent, and the states belonging to a passage class are transient, cf. Section 9.4.
3. [BP94, Theorem (3.9), Chap. 8] Let P be the transition matrix of a finite Markov chain. The Markov chain is
 - (a) Positive recurrent if and only if P is irreducible.
 - (b) Regular if and only if P is primitive.
 - (c) Periodic if and only if P is irreducible and periodic.
4. [BP94, Theorem (3.16), Chap. 8] Let P be the transition matrix of a finite irreducible Markov chain. Then the Markov chain is periodic if and only if P is a weakly cyclic matrix.
5. If P is the transition matrix of a regular Markov chain, then there exists $\lim_{n \rightarrow \infty} P^n = \mathbf{1}\pi$, where π is the probability invariant vector. If the Markov chain is periodic, the sequence $\{P^n\}_{n \geq 0}$ is bounded, but not convergent.
6. [Mey89] Assume that $E = \{1, 2, \dots, k\}$ and that P is irreducible. Let $1 \leq m \leq k - 1$ and $\alpha = \{1, \dots, m\}$, $\beta = \{m + 1, \dots, k\}$. Then
 - (a) The matrix $I - P[\alpha]$ is an M -matrix.
 - (b) The matrix $P' = P[\alpha] + P[\alpha, \beta](I - P[\beta])^{-1}P[\beta, \alpha]$, such that $I - P'$ is the Schur complement of $I - P[\beta]$ in the matrix

$$\begin{bmatrix} I - P[\alpha] & -P[\alpha, \beta] \\ -P[\beta, \alpha] & I - P[\beta] \end{bmatrix},$$

is a stochastic irreducible matrix.

- (c) If we partition the invariant probability vector π as $\pi = [\pi_\alpha, \pi_\beta]$, with $\pi_\alpha = (\pi_i)_{i \in \alpha}$ and $\pi_\beta = (\pi_i)_{i \in \beta}$, we have $\pi_\alpha P' = \pi_\alpha$, $\pi_\beta = \pi_\alpha P[\alpha, \beta](I - P[\beta])^{-1}$.

Examples:

1. The matrix

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

is the transition matrix of a periodic Markov chain of period 2. In fact, P is an irreducible matrix of period 2.

2. The Markov chain with transition matrix

$$P = \begin{bmatrix} \frac{1}{5} & \frac{1}{5} \\ 1 & 0 \end{bmatrix}$$

is regular. In fact, $P^2 > 0$, i.e., P is primitive.

3. Let

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

be the transition matrix of a Markov chain. The matrix P is irreducible and aperiodic, therefore the Markov chain is regular. The vector $\pi = \frac{1}{13}[4, 4, 3, 2]$ is the stationary distribution. According to Fact 5, one has

$$\lim_{n \rightarrow \infty} P^n = \frac{1}{13} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 4 & 4 & 3 & 2 \end{bmatrix}.$$

54.5 Censoring

Definitions:

Partition the state space E into two disjoint subsets, α and β , and denote by $\{t_0, t_1, t_2, \dots\}$ the time when the Markov chain visits the set α :

$$t_0 = \min\{n \geq 0 : X_n \in \alpha\}, \quad t_{k+1} = \min\{n \geq t_k + 1 : X_n \in \alpha\},$$

for $k \geq 0$. The **censored process** restricted to the subset α is the sequence $\{Y_n\}_{n \geq 0}$, where $Y_n = X_{t_n}$, of successive states visited by the Markov chain in α .

Facts:

1. [BLM05, Sec. 1.6] The censored process $\{Y_n\}$ is a Markov chain.
2. [BLM05, Sec. 1.6] Arrange the states so that the transition matrix can be partitioned as

$$P = \begin{bmatrix} P[\alpha] & P[\alpha, \beta] \\ P[\beta, \alpha] & P[\beta] \end{bmatrix}.$$

Then the transition matrix of $\{Y_n\}$ is

$$P' = P[\alpha] + \sum_{n=0}^{+\infty} (P[\alpha, \beta] P[\beta]^n P[\beta, \alpha])$$

provided that $\sum_{n=0}^{+\infty} (P[\alpha, \beta] P[\beta]^n P[\beta, \alpha])$ is convergent. If the series $S' = \sum_{n=0}^{+\infty} P[\beta]^n$ is convergent, then we may rewrite P' as $P' = P[\alpha] + P[\alpha, \beta] S' P[\beta, \alpha]$.

3. [BLM05, Theorem 1.23] Assume that the Markov chain is irreducible and positive recurrent. Partition the stationary distribution π as $\pi = [\pi_\alpha, \pi_\beta]$, with $\pi_\alpha = (\pi_i)_{i \in \alpha}$ and $\pi_\beta = (\pi_i)_{i \in \beta}$. Then one has $\pi_\alpha P' = \pi_\alpha$, $\pi_\beta = \pi_\alpha P[\alpha, \beta] S'$, where P' and S' are defined in Fact 2.
4. [BLM05, Secs. 1.6, 3.5, 4.5] In the case of finite Markov chains, censoring is equivalent to Schur complementation (see Fact 6 of Section 54.4). Censoring is at the basis of several numerical methods for computing the invariant probability vector π ; indeed, from Fact 3, if the invariant probability vector π_α associated with the censored process is available, the vector π_β can be easily computed from π_α . A smart choice of the sets α and β can lead to an efficient computation of the stationary distribution. As an example of this fact, Ramaswami's recursive formula for computing the vector π associated with an M/G/1-type Markov chain is based on successive censorings, where the set A is finite, and $\beta = \mathbb{N} - \alpha$.

54.6 Numerical Methods

Definitions:

Let A be a $k \times k$ matrix. A splitting $A = M - N$, where $\det M \neq 0$, is a **regular splitting** if $M^{-1} \geq 0$ and $N \geq 0$. A regular splitting is said to be **semiconvergent** if the matrix $M^{-1}N$ is semiconvergent.

Facts:

1. The main computational problem in Markov chains is the computation of the invariant probability vector π . If the space state E is finite and the Markov chain is irreducible, classical techniques for solving linear systems can be adapted and specialized to this purpose. We refer the reader to the book [Ste94] for a comprehensive treatment on these numerical methods; in facts below we analyze the methods based on LU factorization and on regular splittings of M_0 -matrices. If the space state E is infinite, general numerical methods for computing π can be hardly designed. Usually, when the state space is infinite, the matrix P has some structures which are specific of the real problem modeled by the Markov chain, and numerical methods for computing π which exploit the structure of P can be designed. Quite common structures arising in queueing problems are the (block) tridiagonal structure, the (block) Hessenberg structure, and the (block) Toeplitz structure (see, for instance, Example 4 Section 54.1). We refer the reader to the book [BLM05] for a treatment on the numerical solution of Markov chains, where the matrix P is infinite and structured.
2. [BP94, Cor. (4.17), Chap. 6], [Ste94, Sec. 2.3] If P is a $k \times k$ irreducible stochastic matrix, then the matrix $I - P$ has a unique LU factorization $I - P = LU$, where L is a lower triangular matrix with unit diagonal entries and U is upper triangular. Moreover, L and U are M_0 -matrices, U is singular, $U\mathbf{1}_k = 0$, and $(U)_{k,k} = 0$.
3. [Ste94, Sec. 2.5] The computation of the LU factorization of an M_0 -matrix by means of Gaussian elimination involves additions of nonnegative numbers in the computation of the off-diagonal entries of L and U , and subtractions of nonnegative numbers in the computation of the diagonal entries of U . Therefore, numerical cancellation cannot occur in the computation of the off-diagonal entries of L and U . In order to avoid possible cancellation errors in computing the diagonal entries of U , Grassmann, Taksar, and Heyman have introduced in [GTH85] a simple trick, which fully exploits the property that U is an M_0 -matrix such that $U\mathbf{1}_k = 0$. At the general step of the elimination procedure, the diagonal entries are not updated by means of the classical elimination formulas, but are computed as minus the sum of the off-diagonal entries. Details are given in Algorithm 1.
4. If $I - P = LU$ is the LU factorization of $I - P$, the invariant probability vector π is computed by solving the two triangular systems $\mathbf{y}L = \mathbf{x}$ and $\mathbf{x}U = 0$, where $\mathbf{x}$ and $\mathbf{y}$ are row vectors, and then by normalizing the solution $\mathbf{y}$. From Fact 2, the nontrivial solution of the latter system is $\mathbf{x} = \alpha \mathbf{e}_k^T$, for any $\alpha \neq 0$. Therefore, if we choose $\alpha = 1$, the vector π can be simply computed by solving the system $\mathbf{y}L = \mathbf{e}_k^T$ by means of back substitution, and then by setting $\pi = (1/\mathbf{y}\mathbf{1})\mathbf{y}$, as described in Algorithm 2.
5. [BP94, Cor. (4.17), Chap. 6] If P is a $k \times k$ irreducible stochastic matrix, the matrix $I - P^T$ has a unique LU factorization $I - P^T = LU$, where L is a lower triangular matrix with unit diagonal entries, and U is upper triangular. Moreover, both L and U are M_0 -matrices, U is singular, and $(U)_{k,k} = 0$. Since L is nonsingular, the invariant probability vector can be computed by calculating a nonnegative solution of the system $U\mathbf{y} = 0$ by means of back substitution, and by setting $\pi = (1/\|\mathbf{y}\|_1)\mathbf{y}^T$.
6. If P is a $k \times k$ irreducible stochastic matrix, and if $\hat{P}$ is the $(k-1) \times (k-1)$ matrix obtained by removing the i th row and the i th column of P , where $i \in \{1, \dots, k\}$, then the matrices $I - \hat{P}^T$ and $I - \hat{P}$ have a unique LU factorization, where both the factors L and U are M -matrices. Therefore, if $i = k$ and if the matrix $I - P^T$ is partitioned as

$$I - P^T = \begin{bmatrix} I - \hat{P}^T & \mathbf{a} \\ \mathbf{b}^T & c \end{bmatrix},$$

one finds that the vector $\hat{\pi} = [\pi_1, \dots, \pi_{k-1}]^T$ solves the nonsingular system $(I - \hat{P}^T)\hat{\pi} = -\pi_k \mathbf{a}$. Therefore, the vector π can be computed by solving the system $(I - \hat{P}^T)\mathbf{y} = -\mathbf{a}$, say by means of LU factorization, and then by setting $\pi = [\mathbf{y}^T, 1]/(1 + \|\mathbf{y}\|_1)$.

7. [Ste94, Sec. 3.6] Let P be a finite irreducible stochastic matrix, and let $I - P^T = M - N$ be a regular splitting of $I - P^T$. Then the matrix $H = M^{-1}N$ has spectral radius 1 and 1 is a simple eigenvalue of H .
8. [Ste94, Theorem 3.3], [BP94], [Var00] Fact 7 above is not sufficient to guarantee that any regular splitting of $I - P^T$, where P is a finite stochastic irreducible matrix, is semiconvergent (see the Example below). In order to be semiconvergent, the matrix H must not have eigenvalues of modulus 1 different from 1, and the Jordan blocks associated with 1 must have dimension 1.
9. An iterative method for computing π based on a regular splitting $I - P^T = M - N$ consists in generating the sequence $\mathbf{x}_{n+1} = M^{-1}N\mathbf{x}_n$, for $n \geq 0$, starting from an initial column vector $\mathbf{x}_0$. If the regular splitting is semiconvergent, by choosing $\mathbf{x}_0 \geq 0$ such that $\|\mathbf{x}_0\|_1 = 1$, the sequence $\{\mathbf{x}_n\}_n$ converges to π^T . The convergence is linear and the asymptotic rate of convergence is the second largest modulus eigenvalue θ of $M^{-1}N$.
10. [Ste94, Theorem 3.6] If P is a finite stochastic matrix, the methods of Gauss–Seidel, Jacobi, and SOR, for $0 < \omega \leq 1$, applied to $I - P^T$, are based on regular splittings.
11. If E is finite, the invariant probability vector is, up to a multiplicative constant, the nonnegative left eigenvector associated with the dominating eigenvalue of P . The power method applied to the matrix P^T is the iterative method defined by the trivial regular splitting $I - P^T = M - N$, where $M = I$, $N = P^T$. If P is primitive, the power method is convergent.
12. [Ste94, Theorems 3.6, 3.7] If P is a $k \times k$ irreducible stochastic matrix, and if $\hat{P}$ is the $(k-1) \times (k-1)$ matrix obtained by removing the i th row and the i th column of P , where $i \in \{1, \dots, k\}$, then any regular splitting of the matrix $I - \hat{P}^T = M - N$ is convergent. In particular, the methods of Gauss–Seidel, Jacobi, and SOR, for $0 < \omega \leq 1$, applied to $I - \hat{P}^T$, being based on regular splittings, are convergent.
13. [Ste94, Theorem 3.17] Let P be a $k \times k$ irreducible stochastic matrix and let $\epsilon > 0$. Then the splitting $I - P^T = M_\epsilon - N_\epsilon$, where $M_\epsilon = D - L + \epsilon I$ and $N_\epsilon = U + \epsilon I$, is a semiconvergent regular splitting for every $\epsilon > 0$.
14. [Ste94, Theorem 3.18] Let P be a $k \times k$ irreducible stochastic matrix, let $I - P^T = M - N$ be a regular splitting of $I - P^T$, and let $H = M^{-1}N$. Then the matrix $H_\alpha = (1 - \alpha)I + \alpha H$ is semiconvergent for any $0 < \alpha < 1$.

Algorithms:

1. The following algorithm computes the LU factorization of $I - P$ by using the Grassmann, Taksar, Heyman (GTH) trick of Fact 3:

Computation of the LU factorization of $I - P$ with GTH trick

Input: the $k \times k$ irreducible stochastic matrix P .

Output: the matrix $A = [a_{i,j}]$ such that $a_{i,j} = l_{i,j}$ for $i > j$, and $a_{i,j} = u_{i,j}$ for $i \leq j$, where $L = [l_{i,j}]$, $U = [u_{i,j}]$ are the factors of the LU factorization of $I - P$.

Computation:

- 1– set $A = I - P$;
- 2– for $m = 1, \dots, k - 1$:
 - (a) for $i = m + 1, \dots, k$, set $a_{i,m} = a_{i,m}/a_{m,m}$;
 - (b) for $i, j = m + 1, \dots, k$, set $a_{i,j} = a_{i,j} - a_{i,m}a_{m,j}$ if $i \neq j$;
 - (c) for $i = m + 1, \dots, k$ set $a_{i,i} = -\sum_{l=m+1, l \neq i}^k a_{l,i}$.

2. The algorithm derived by Fact 4 above is the following:

Computation of π through LU factorization

Input: the factor $L = [l_{i,j}]$ of the LU factorization of the matrix $I - P$, where P is a $k \times k$ irreducible stochastic matrix.

Output: the invariant probability vector $\pi = (\pi_i)_{i=1,k}$.

Computation:

- 1– set $\pi_k = 1$;
- 2– for $j = k - 1, \dots, 1$ set $\pi_j = -\sum_{i=j+1}^k \pi_i l_{i,j}$;
- 3– set $\pi = (\sum_{i=1}^k \pi_i)^{-1} \pi$.

Examples:

The matrix

$$P = \begin{bmatrix} 0.5 & 0 & 0 & 0.5 \\ 0.2 & 0.8 & 0 & 0 \\ 0 & 0.6 & 0.4 & 0 \\ 0 & 0 & 0.1 & 0.9 \end{bmatrix}$$

is the transition matrix of an irreducible and aperiodic Markov chain. The splitting $I - P^T = M - N$ associated with the Jacobi method is a regular splitting. One may easily verify that the iteration matrix is

$$M^{-1}N = D \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} D^{-1},$$

where $D = \text{diag}(1, \frac{5}{2}, \frac{5}{6}, 5)$. Therefore, the regular splitting is not semiconvergent since the iteration matrix has period 4 (see Fact 8 above).

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